

RESEARCH DOSSIER AND IMPLEMENTATION PLAN

Cross-Feature Fusion Mamba Networks for Streaming Small-Object Detection

*A YOLO-based, state-space, visible–thermal detector,
from still images to live camera streams*

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This document consolidates a literature review, a reading of four source papers, a survey of work published up to 2 October 2026, and a plan for building and evaluating the system it describes. Figures and tables that report results from other work reproduce numbers stated in the cited papers or their official repositories. Where a figure could not be confirmed against a primary source, it is marked with a dagger ([†]) and listed in Appendix B. All targets stated for the proposed method are hypotheses to be tested, not results.

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Summary

This dossier turns the supervised project topic *YOLO-Based Cross Feature Fusion Mamba Net for Object Detection* into a research programme that can support a journal article. It has three parts: an evidence base, a proposed method, and an implementation plan.

EVIDENCE BASE. The four source papers were read in full, including their tables and reference lists: MambaVision [51], HPS-DETR [145], UAVDet [156] and CFMW [69]. They were then placed against work published up to October 2026. Three findings shape everything that follows.

- On still images, most of the accuracy gain these papers report comes from familiar devices. These are a stride-4 detection head, more capacity in early layers, and better multi-scale fusion. The state-space (Mamba) component itself adds between +0.6 and about +3 points of mAP, often at a large cost in speed or computation.
- Combining a YOLO detector with a Mamba fusion block for visible and thermal images is already well covered. Fusion-Mamba [29], COMO [14], WaveMamba [148], MS2Fusion [91] and more than twenty Mamba-in-YOLO variants occupy that space. A further static fusion block would be judged incremental.
- No published method was found that carries a state-space memory causally across an unbounded stream of visible–thermal frames, at constant memory, for small objects. The nearest work, MambaST [86], uses three-frame windows on one pedestrian dataset and does not evaluate streaming.

PROPOSED METHOD. CFFM-Net (Cross-Feature Fusion Mamba Network) keeps the supervised topic intact. It is a dual-stream YOLO detector with a stride-4 small-object head and state-space fusion of visible and thermal features. Its streaming variant, CFFM-Net-S, adds a *Temporal State Memory* that carries the state-space hidden state from frame to frame, after correcting for camera motion. Both variants rest on one observation. In a selective state-space

model the discretisation step Δ decides whether the state keeps its memory or takes in new input. CFFM-Net uses this single control for three purposes: to discount an unreliable modality at a location, to carry objects through gaps in a sensor feed, and to adjust for irregular frame intervals. Five hypotheses with explicit falsification criteria are stated in Section 6.9.

IMPLEMENTATION PLAN. The system is built in four phases, each a usable product in its own right.

1. Phase 1 detects objects in still images from curated datasets.
2. Phase 2 detects and tracks objects in uploaded video.
3. Phase 3 runs on live webcam and network camera streams.
4. Phase 4, an optional extension, adds anonymous multi-camera analytics.

The core benchmark is RGBT-Tiny [157]. It contains 115 paired visible–thermal sequences, and more than 81% of its objects are smaller than 16×16 pixels. A notebook-driven workflow of sixteen notebooks runs over a tested Python package. Training runs on Kaggle T4 GPUs, while a local RTX 4050 laptop GPU is used for development and live demonstrations.

ETHICS AND LAW. The fictional Machine of *Person of Interest* motivated the project, but the system deliberately stops short of it. It does not identify people, keep watch-lists or profile behaviour. Those functions are prohibited or high-risk under the EU AI Act [36] and constrained by India’s Digital Personal Data Protection Act [30]. The system is a perception layer that detects and tracks objects and anonymous people, with faces blurred when footage is ingested.

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Introduction

1.1 The question behind the project

In the television series *Person of Interest* [101], a system called the Machine watches the cameras of a city. It recognises people, follows them from one camera to the next and predicts which of them will be involved in violence. As fiction it combines several technologies that are real today and several that are not. Its lowest layer is the one this project is about: a perception system that looks at camera feeds and says, reliably and in real time, what objects are present and where they are.

The supervised project topic, *YOLO-Based Cross Feature Fusion Mamba Net for Object Detection*, names three ingredients of such a layer. YOLO is the family of single-stage detectors that made real-time detection practical on ordinary hardware [132, 144, 160]. Cross-feature fusion is the combination of information from different sources, such as a visible-light camera and a thermal camera, or features taken at different image scales. Mamba is a selective state-space model, a sequence model whose cost grows linearly with sequence length and which summarises everything it has seen in a fixed-size hidden state [48].

This dossier sets out how these ingredients can be combined into work that is new, measurable and publishable, and how the result can be built in phases that end with a system running on live cameras.

1.2 From a fictional Machine to a defensible research problem

A research contribution needs a precise problem, a measurable claim and an honest account of prior work. The Machine is none of these things. Three choices narrow it into something that can be studied.

The first choice is the scene. Objects that occupy few pixels are the hard case for every detector. A person 200 metres from a camera or a car seen from a drone may cover fewer than 16×16 pixels. Small-object detection remains the weakest part of the COCO benchmark [70]. It has dedicated benchmarks, including VisDrone [171], AI-TOD [154], SODA [13] and RGBT-Tiny [157].

The second choice is the sensor. Visible cameras fail at night, in glare and in fog. Thermal cameras see heat regardless of light but lose colour and texture. A detector that fuses both is useful exactly when a single sensor is not [29, 69].

The third choice is time. A camera does not produce isolated photographs. It produces a stream in which a faint object seen in one frame is usually still there in the next. Most published detectors, including all four source papers studied here, process each image independently. State-space models were designed for sequences, so carrying their memory across frames is a natural use of them. As Chapter 5 shows, it is also the least explored one.

1.3 Data in, detections out

A common early question is whether the system works only on a dataset that is handed to it, or whether it can also watch a live camera. It can do both, and the distinction is worth stating precisely because it organises the whole plan.

Training always uses curated datasets. These are collections of images or video frames in which every object of interest has been labelled with a box and a class. The model learns from these labels, and its accuracy is measured on a held-out portion of the same datasets. This part of the work is the same whatever the final application.

Inference, the use of the trained model, can draw on any source of images. The same weights can read a folder of test images, an uploaded video file or a live stream from a webcam, an IP camera or a thermal camera. The detector itself does not change. What changes is the work around it. Video must be decoded and its frames kept in order. A live stream must be read fast enough to keep up, so stale frames must be dropped. A streaming model must also keep its memory from one frame to the next. Figure 1.1 shows the four phases that follow from this.

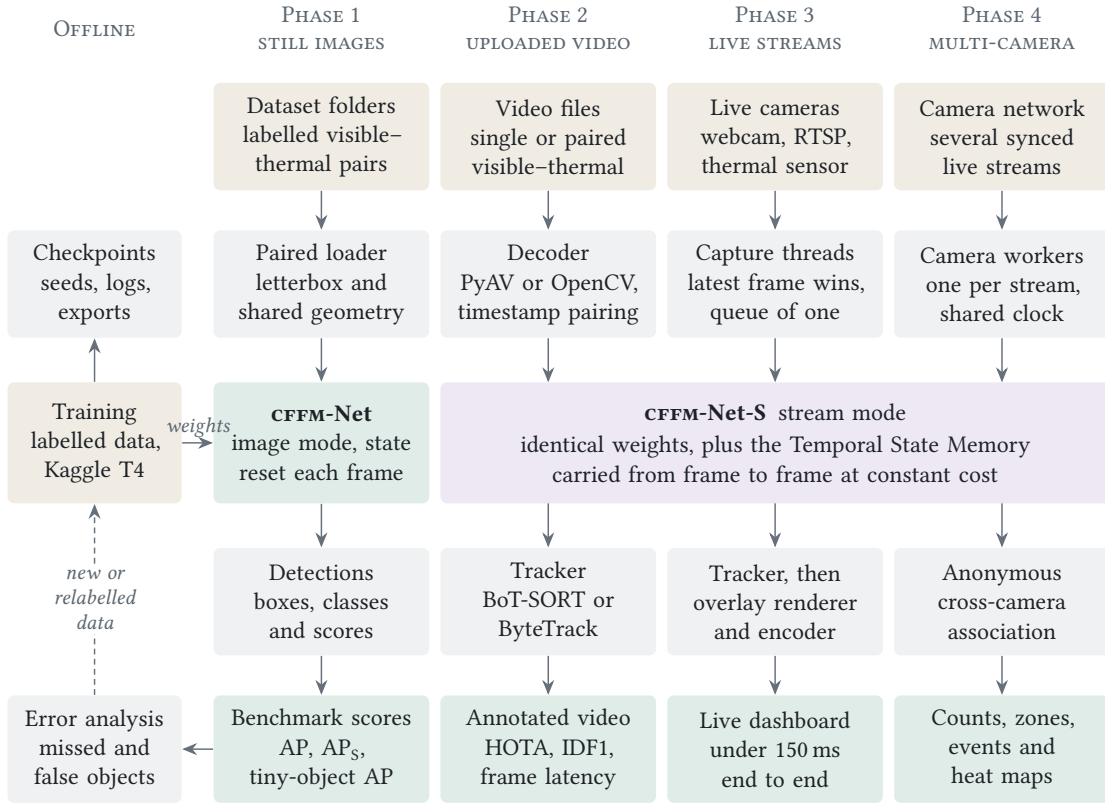


FIGURE 1.1. The system at a glance. Training always uses labelled datasets. The trained weights then serve four kinds of input, one per phase. Phase 1 evaluates the image-mode network on still images. Phases 2 to 4 run the same weights in stream mode, where the Temporal State Memory of **CFFM-Net-S** carries information between frames. Errors found on benchmarks and in the field are analysed and turned into new or corrected training data.

1.4 Research questions

The plan is organised around four questions. Each can be answered by experiment.

- rQ1. Does a state-space memory carried causally across frames improve the detection of very small objects in visible–thermal video, compared with the same detector applied frame by frame?
- rQ2. Is the selective state-space formulation the right form for that memory? Specifically, does it beat a convolutional recurrent cell and windowed temporal attention at matched computation?
- rQ3. Can the discretisation step of the state-space model act as a learned reliability gate? If so, the detector should degrade gracefully when one sensor drops out, drifts out of alignment or delivers frames irregularly.
- rQ4. Can the resulting detector run on live streams within a real-time budget, on a desktop GPU and on an embedded Jetson module, without losing the accuracy measured offline?

1.5 What this dossier contributes

The dossier is a planning document, not a results paper. It contributes the following.

- A critical reading of the four source papers. The reading reconstructs their tables from the original PDFs and identifies where their gains come from (Chapter 3).
- A survey of the state of the art as of October 2026, newest first, in six areas. These are real-time detectors, vision state-space models, small objects, visible–thermal fusion, video and tracking, and foundation models (Chapter 4).
- A novelty analysis of 64 closely related papers. It shows which directions are saturated and which remain open (Chapter 5).
- The design of CFFM-Net and CFFM-Net-S, with equations, complexity analysis and falsifiable hypotheses (Chapter 6).
- A dataset stack, an experimental protocol that meets the standards of the target journals, a four-phase roadmap and a notebook workflow (Chapters 7 to 10).
- An account of the legal and ethical limits that separate this system from the fictional Machine (Chapter 11).

1.6 How the programme fits together

Figure 1.2 shows the order of work and how later stages feed back into earlier ones. It also serves as a reading guide.

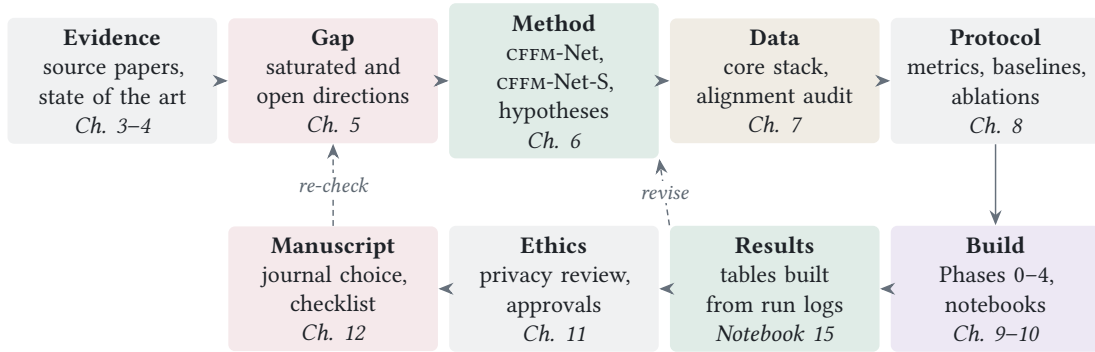


FIGURE 1.2. The research programme. Solid arrows give the order of work and the chapters that describe each stage. The two dashed arrows are the feedback loops that matter most. Ablation results revise the method. Prior art is checked again immediately before submission, because the field publishes new Mamba detectors every month.

1.7 Conventions

Numbers quoted from other work are taken from the cited paper or its official repository. The dagger ([†]) marks figures that could only be confirmed from a secondary source. Dates in tables are the month a work first appeared, usually as an arXiv preprint. Mean average precision at an IoU threshold of 0.5 is written AP_{50} . The COCO average over thresholds 0.5 to 0.95 is written AP or $mAP_{50:95}$. Small-object metrics follow the definitions in Section 2.2.

Technical Foundations

This chapter introduces the ideas the rest of the dossier depends on. It covers how a YOLO detector is organised and why small objects strain it. It then derives the selective state-space model behind Mamba and shows how such models have been adapted to images. It ends with visible–thermal fusion and the step from images to video. Readers familiar with any section can skip it. Section 2.3.1 is the exception, because it contains the observation on which the proposed method is built.

2.1 Real-time detection and the YOLO family

A single-stage detector predicts object boxes and classes in one pass over the image, with no separate proposal stage. The YOLO family has three parts [132, 144, 160]. A *backbone* turns the image into feature maps at progressively coarser resolution. A *neck* mixes information between those resolutions. A *head* predicts, for each location of each output map, whether an object is centred there, its box and its class.

The resolution of a feature map is described by its *stride*, the number of input pixels per feature cell. Following common notation, the map at stride 2^k is called P_k . For a 640×640 input, P_3 has 80×80 cells at stride 8, P_4 has 40×40 at stride 16 and P_5 has 20×20 at stride 32. Standard YOLO models predict from P_3 , P_4 and P_5 . Small-object variants add a P_2 head at stride 4, with 160×160 cells [156, 170].

The neck follows the feature pyramid network [71] and its path-aggregation extension [72]. A top-down path carries semantic information from coarse maps to fine ones. A bottom-up path then returns localisation detail to the coarse maps.

Two further design choices matter for this project. The first is *label assignment*, the rule that decides which predictions are trained to match which ground-truth objects. Tiny objects cover so few cells that a poor rule may give them no positive samples at all. YOLO26 introduces a small-target-aware assignment for this reason [160]. The second is *non-maximum suppression* (NMS), which removes duplicate boxes after prediction. NMS costs time that papers often leave out of latency figures [111]. YOLOv10 and YOLO26 offer heads trained with one-to-one matching that need no NMS [144, 160]. Figure 2.1 summarises the layout and the numbers that recur in later chapters.

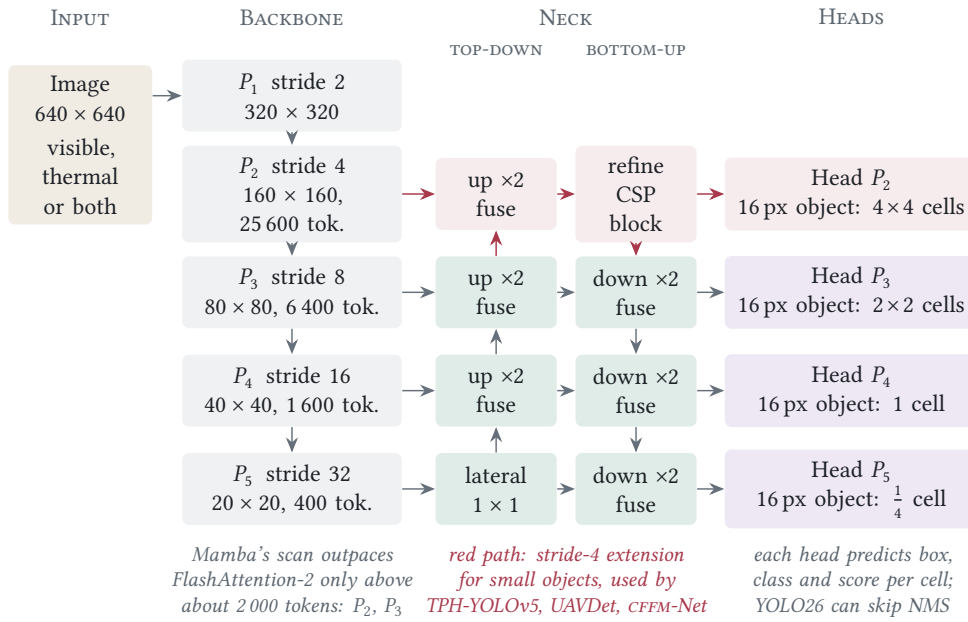


FIGURE 2.1. Anatomy of a YOLO detector at 640×640 input. The backbone halves resolution at each stage. The neck carries semantics down the pyramid (top-down) and localisation back up (bottom-up). Heads predict at each retained level. The red path is the stride-4 extension used for small objects. The head labels show how many cells a 16-pixel object covers at each level, and the backbone labels give the token count an SSM would scan.

2.2 What makes an object small

Size can be defined in absolute terms, by the number of pixels a box covers, or relative to the image. Absolute definitions dominate, but the thresholds differ between benchmarks (Table 2.1). The COCO thresholds are the most widely reported. They are too coarse for aerial and long-range imagery, which is why AI-TOD and SODA subdivide the range below 32 pixels.

TABLE 2.1. Definitions of small and tiny objects used by the benchmarks in this dossier. Sizes refer to the square root of box area in pixels unless stated otherwise.

Benchmark	Size bands	Notes
COCO [70]	small < 32; medium 32–96; large > 96	AP_S, AP_M, AP_L
AI-TOD [142, 154]	very tiny 2–8; tiny 8–16; small 16–32; medium 32–64	mean object size about 12.8 px
SODA-D, SODA-A [13]	extremely small ≤ 12 ; relatively small ≤ 20 ; generally small $\leq 32^\dagger$	driving and aerial scenes
TinyPerson [162]	persons averaging about 18 px	ignore regions are common
RGBT-Tiny [157]	81% of objects below 16×16	proposes the SAFit matching measure
CFMW (SWVI) [69]	small if area < 2500 px^2 at 1280×1024	26.2% of objects are small

Three mechanisms make small objects hard. The first is *sampling*. A 16-pixel object covers half a cell of P_5 and four cells of P_3 . Each stride-2 downsampling step discards information that later layers cannot recover. The second is *matching*. Intersection over union (IoU) is unforgiving at small scales. A box offset by four pixels can drop a 16-pixel object from IoU 1.0 to below 0.5. The normalised Wasserstein distance was proposed to soften this [143]. The third is *context*. A faint blob may be a pedestrian or noise, and the surrounding scene often decides which. Global context helps here, and so does temporal context: the same blob seen moving consistently over several frames. Temporal context is the lever that Chapter 6 pulls.

2.3 State-space models, from S4 to Mamba

A linear state-space model maps an input signal $x(t)$ to an output $y(t)$ through a hidden state $h(t) \in \mathbb{R}^N$:

$$h'(t) = A h(t) + B x(t), \quad y(t) = C h(t) + D x(t). \quad (2.1)$$

To apply it to a sequence $x_1, \dots, x_L$ sampled with step Δ , the system is discretised. Under the zero-order hold,

$$\bar{A} = \exp(\Delta A), \quad \bar{B} = (\Delta A)^{-1}(\exp(\Delta A) - I) \Delta B \approx \Delta B, \quad (2.2)$$

which gives the recurrence

$$h_k = \bar{A} h_{k-1} + \bar{B} x_k, \quad y_k = C h_k + D x_k. \quad (2.3)$$

When A , B , C and Δ do not depend on the input, the output equals a convolution of the input with the kernel $\bar{K} = (C\bar{B}, C\bar{A}\bar{B}, \dots, C\bar{A}^{L-1}\bar{B})$. Structured state-space models such as S4 exploit this to train in parallel and run recurrently [47, 126].

Mamba makes B , C and Δ functions of the current input,

$$B_k = W_B x_k, \quad C_k = W_C x_k, \quad \Delta_k = \text{softplus}(w_\Delta^\top x_k + b_\Delta), \quad (2.4)$$

so that the model can choose, token by token, what to store and what to ignore [48]. The convolutional view is lost. A hardware-aware parallel scan recovers training speed, and inference remains a recurrence with a fixed-size state. Cost is linear in sequence length, against the quadratic cost of self-attention. Mamba-2 later recast the model as a structured form of linear attention [19]. An independent analysis reached the same view and found that Mamba’s forget gate and block design account for most of its advantage [50].

2.3.1 The discretisation step as a gate

In Mamba, A is diagonal with negative real entries. For each channel the transition $\bar{A}_k = \exp(\Delta_k A)$ therefore lies between 0 and 1, and Δ_k behaves like a gate.

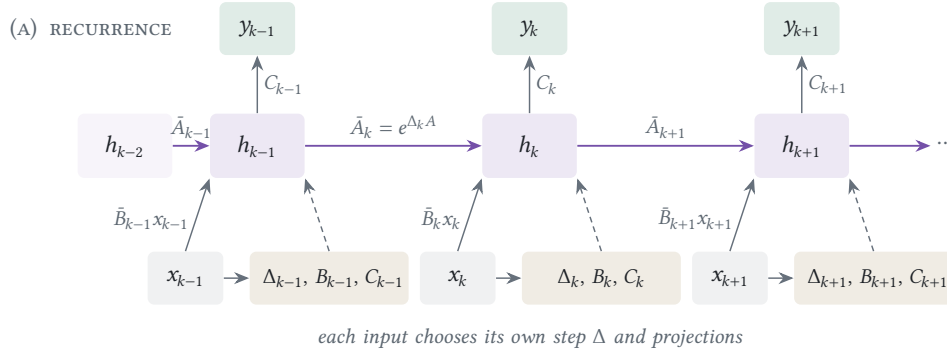
- As $\Delta_k \rightarrow 0$, $\bar{A}_k \rightarrow I$ and $\bar{B}_k \rightarrow 0$. The state is kept unchanged and the input is ignored.
- As $\Delta_k \rightarrow \infty$, $\bar{A}_k \rightarrow 0$. The past is forgotten and the state is rebuilt from the current input.

Gu and Dao make the link precise. With a one-dimensional state, $A = -1$ and $B = 1$, the selective recurrence reduces to the gated update $h_k = (1 - g_k) h_{k-1} + g_k x_k$ with $g_k = \sigma(\text{Linear}(x_k))$ (Theorem 1 of [48]). Figure 2.2 shows the recurrence and the two regimes.

This reading has a second consequence. Equation (2.1) is a continuous-time system, and Δ is literally a time step. If samples arrive at irregular intervals, scaling Δ by the true interval keeps the discrete model faithful to the same continuous dynamics. Zubić and colleagues used this property to make event-camera detectors generalise across sampling frequencies [174]. The proposed method uses one control, Δ , for three jobs: weighting sensor reliability, bridging gaps in a stream and adjusting for frame timing.

2.4 State-space models for images

An image is not a sequence, so an SSM must be given an order in which to read it. The order matters, because a one-dimensional causal scan places pixels that are adjacent in two dimensions far apart in time. Vision Mamba (Vim) scans the patch sequence forwards and



(B) THE ROLE OF Δ

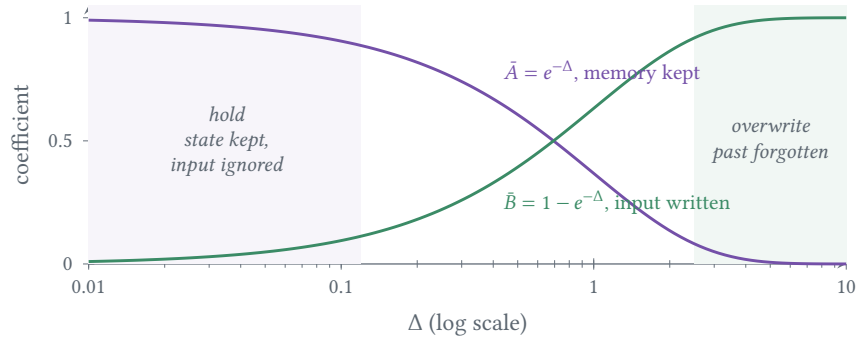


FIGURE 2.2. The selective state-space recurrence. (a) Each input x_k produces its own step Δ_k and projections B_k and C_k . The state h is carried forward through $\bar{A}_k = \exp(\Delta_k A)$. (b) For a scalar state with $A = -1$ and $B = 1$, the zero-order-hold coefficients depend only on Δ . A small Δ holds the state and ignores the input. A large Δ forgets the past and writes the input. CFM-Net exploits this single control to discount unreliable sensors and to bridge gaps in a stream.

backwards [172]. VMamba’s SS2D unrolls the feature map in four directions and merges the results [76]. LocalMamba scans within windows [57]. In 2025 and 2026, Spatial-Mamba [151], DAMamba [17], PRISMamba [102], VNCT [140] and GraphScan [46] added adaptive and non-directional alternatives. VNCT goes so far as to argue that directional scanning is unnecessary. Figure 2.3 shows the main families.

MambaVision is one of the source papers, analysed in Section 3.1. It takes a hybrid route [51]. Its first two stages are convolutional. Its last two alternate a modified Mamba mixer with self-attention, with attention placed in the later layers.

Two results temper enthusiasm for vision SSMs and must be addressed by any paper that uses them. MambaOut removed the state-space component from vision Mamba models and found the result *better* on ImageNet classification [163]. It still trailed VMamba by 1.4 box AP on COCO detection, so the SSM may earn its place on dense tasks. Second, Mamba’s scan is faster than FlashAttention-2 only beyond about two thousand tokens [18, 48]. At 640×640 input, P_5 has 400 tokens and P_4 has 1 600. At those levels, attention or plain convolution is usually faster in practice, as SF-Mamba reports for short vision sequences [120]. Both points shape where CFFM-Net places its state-space blocks.

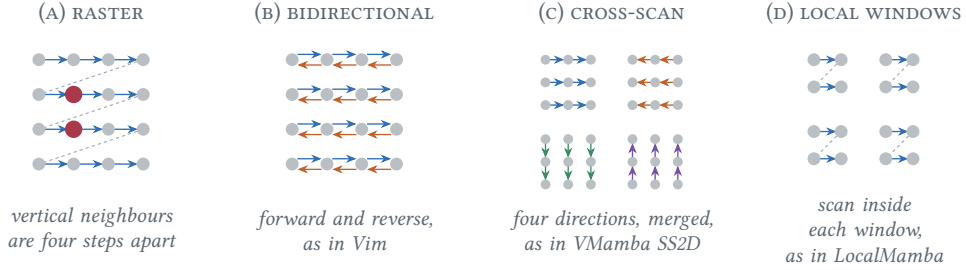


FIGURE 2.3. Scan orders used to apply state-space models to images, shown on small grids of feature cells. (a) A raster scan places vertical neighbours (red) a full row apart in sequence order. (b) Vim adds a reverse pass. (c) VMamba’s SS2D scans rows and columns in both directions and merges the four outputs. (d) Local scans keep neighbourhoods together. Work in 2026 replaces fixed orders with adaptive or non-directional ones (Section 2.4).

2.5 Visible–thermal fusion

Visible (RGB) cameras record reflected light, and thermal cameras record emitted long-wave infrared. Their failure modes are nearly complementary. RGB is rich in texture and colour but fails in darkness, glare, smoke and fog. Thermal is indifferent to lighting but flat in texture, and it can confuse objects that share a temperature with their background. Fusion methods are usually grouped by where the two streams meet (Figure 2.4):

- *early* or pixel-level fusion stacks the images and feeds one network;
- *middle* or feature-level fusion runs two backbones and exchanges information between their feature maps;
- *late* or decision-level fusion runs two detectors and merges their boxes [11].

Feature-level fusion dominates current work. It began with CNN and attention blocks such as CFT [105] and ICAFusion [122]. State-space blocks followed: CFMW [69], Fusion-Mamba [29], COMO [14], WaveMamba [148] and MS2Fusion [91].

Three practical problems recur.

- *Alignment.* The two sensors rarely share an optical axis, so their images are offset, and the offset varies with depth. KAIST is known for weak alignment [59, 66]. DroneVehicle is only roughly registered [131]. FLIR ADAS is not registered at all, which is why most papers use an aligned subset [165].
- *Label quality.* Labels drawn on one modality do not always fit the other, and some benchmarks have been relabelled for this reason.
- *Modality imbalance.* On LLVIP, a thermal-only YOLO comes within about one to two points of most fusion methods (Section 3.4). Fusion results on that dataset therefore say little.

2.6 From images to video

Video adds a time axis and three research traditions.

- *Video object detection* improves per-frame detection by aggregating features from neighbouring frames. ImageNet VID is its standard benchmark [114]. Most methods use a window of past and future frames, which is acceptable offline but not in a live stream. TMambaDet (CVPR 2026) is a recent example, using a bidirectional temporal Mamba [134].
- *Streaming perception* recognises that the world moves while the detector computes. It scores predictions against the state of the world at the moment they are delivered [67]. StreamYOLO, LongShortNet and DAMO-StreamNet forecast object motion to compensate [53, 68, 155].
- *Multi-object tracking* links detections into identities over time. Tracking-by-detection methods associate boxes with motion and appearance cues: SORT-family trackers,

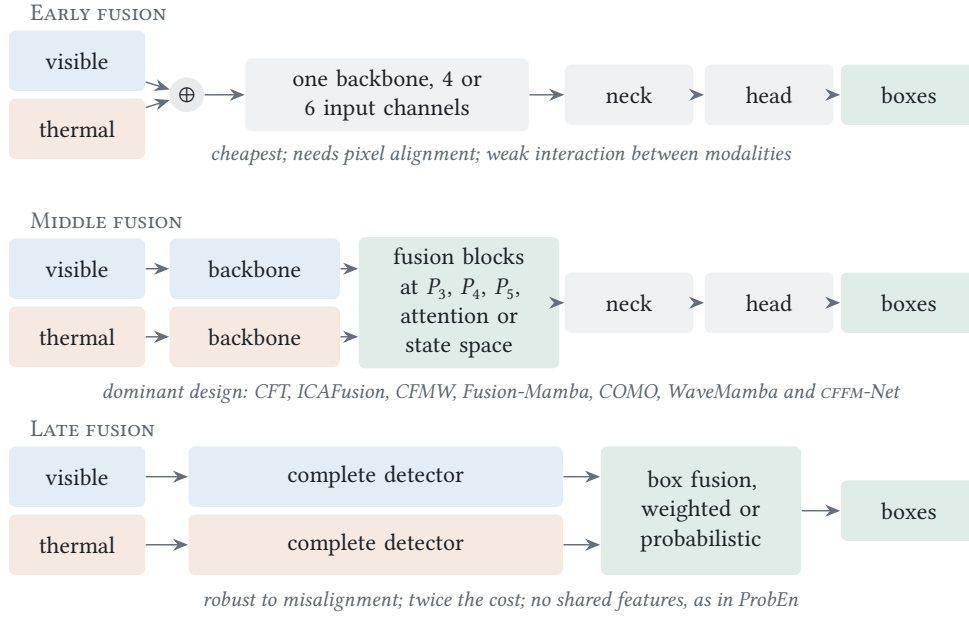


FIGURE 2.4. Three places where visible and thermal streams can meet. Middle fusion exchanges information between feature maps at several scales and dominates current work. CFFM-Net is a middle-fusion design whose fusion blocks are state-space models with reliability-gated steps (Chapter 6).

ByteTrack, BoT-SORT, OC-SORT and TrackTrack [2, 6, 123, 166]. End-to-end trackers carry learned queries from frame to frame: MOTR, MOTIP and the state-space SambaMOTR [41, 118, 168]. Accuracy is measured with HOTA [77] and IDF1 [109].

A causal detector, one that uses only the past, is the natural fit for a live camera. A recurrent state is the most economical way to give it memory. Recurrent detectors exist for event cameras, using recurrent vision transformers [43] and state-space models [174]. Chapter 5 shows that the equivalent for visible–thermal video of small objects has not been published.

Critical Analysis of the Four Source Papers

The project began from four papers supplied with the topic. This chapter reads each in turn and then asks what they show together. The analysis used the full text and reference lists. Because text extraction lost most of their tables, every table was read again from the rendered PDF pages. Where a paper’s text and tables disagree, both values are reported and the disagreement is listed in Table 3.7.

One identification matters from the start. The fourth file, supplied as *Cross-Modality Fusion Mamba for Object Detection*, is CFMW by Li and colleagues [69]. It is not the similarly named Fusion-Mamba by Dong and colleagues [29]. Both are cited throughout, and both belong to the same family of designs. Table 3.1 summarises the four papers.

TABLE 3.1. The four source papers at a glance.

Paper	Venue	Base detector	Task and data	Code
MambaVision [51]	CVPR 2025 (NVIDIA)	Cascade Mask R-CNN	backbone; ImageNet, COCO, ADE20K	public
HPS-DETR [145]	IEEE TGRS 63, 2025	RT-DETR-r18	aerial small objects; SIMD, DOTA, NWPU VHR-10	none
UAVDet [156]	CVIU 264, 2026	RTMDet-m	UAV small objects; Vis-Drone, UAVDT, DroneVehicle (RGB)	none
CFMW [69]	IEEE TCSVT 35(12), 2025	dual-stream YOLOv5-L	visible–thermal under weather; LLVIP, SWVI	announced

3.1 MambaVision: a hybrid Mamba–Transformer backbone

WHAT IT PROPOSES. MambaVision, by Hatamizadeh and Kautz of NVIDIA, is a four-stage hierarchical backbone [51]. Stages one and two, at strides 4 and 8, are plain convolutional residual blocks chosen for speed at high resolution. Stages three and four, at strides 16 and 32, contain N layers each. The first $N/2$ use a redesigned Mamba mixer and the last $N/2$ use windowed self-attention. The mixer departs from Mamba in three ways.

- The causal convolution is replaced by an ordinary one.
- A second branch is added that does not pass through the SSM.
- The multiplicative gate is replaced by concatenating the two branches, followed by a linear projection.

The scan runs once, in one direction.

EVIDENCE. Table 3.2 gives the ImageNet and COCO results that matter here. The ablations support both design choices. Concatenation instead of gating adds 1.0 point of ImageNet top-1 and 1.1 box AP with Mask R-CNN. Placing attention in the last half of each stage gives 82.3% top-1. That beats random placement (81.3%), the first half (81.5%) and alternating patterns (81.4–81.6%).

LIMITATIONS FOR THIS PROJECT.

- The detection gains over ConvNeXt are small: +0.7, +0.4 and +0.1 box AP at the three sizes. They come from single runs with no variance reported.
- No size-specific metric is given, so its effect on small objects is unknown.
- Its state-space and attention blocks work only at strides 16 and 32. A 16-pixel object barely registers at those strides, so the part of the network that should supply global context never sees small objects at useful resolution. This is the central mismatch with the project.
- Throughput is reported only for classification at 224×224 on an A100. No latency is given for a real-time detector, a consumer GPU or batch size one.

WHAT TO TAKE FROM IT. Three lessons carry over. Attention or global mixing belongs late in a stage. A non-SSM branch beside the scan is cheap and helps. The design that would actually serve small objects, cheap global mixing at strides 4 and 8, was not attempted.

TABLE 3.2. MambaVision results reproduced from the paper. Throughput is images per second at 224×224 , batch 128, on an A100. COCO uses Cascade Mask R-CNN with the $3\times$ schedule. Parameters and GFLOPs for COCO refer to the whole detector.

Model	Params (M)	GFLOPs	Img/s	Top-1 (%)	
MambaVision-T	31.8	4.4	6 298	82.3	
MambaVision-S	50.1	7.5	4 700	83.3	
MambaVision-B	97.7	15.0	3 670	84.2	
MambaVision-L	227.9	34.9	2 190	85.0	

COCO backbone	Params (M)	GFLOPs	Box AP	Mask AP	Δ box
ConvNeXt-T	86	741	50.4	43.7	
Swin-T	86	745	50.4	43.7	
MambaVision-T	86	740	51.1	44.3	+0.7
ConvNeXt-S	108	827	51.9	45.0	
MambaVision-S	108	828	52.3	45.2	+0.4
ConvNeXt-B	146	964	52.7	45.6	
MambaVision-B	145	964	52.8	45.7	+0.1

3.2 HPS-DETR: lightweight small-object detection with a transformer

WHAT IT PROPOSES. HPS-DETR, by Wang and Chen, modifies the real-time transformer detector RT-DETR-r18 for small objects in aerial images [145, 169]. It is a method paper, not a survey. It makes four changes.

- A lighter backbone block combines partial convolution from FasterNet [12] with a gated linear unit.
- Cascaded group attention, from EfficientViT [75], replaces standard self-attention in the encoder. It is applied to P_5 only.
- A new fusion neck brings the stride-4 level into scale-sequence and triple-feature fusion. It borrows from ASF-YOLO [61] and adds a learned upsampler derived from DySample [74].
- The box loss combines Inner-IoU and MPDIoU [80, 167].

The parameter count falls from 19.9M to 15.5M.

EVIDENCE. Table 3.3 gives the headline results. The authors repeated every experiment four times and report averages. That is better practice than the other three papers follow.

TABLE 3.3. HPS-DETR results from the paper, baseline RT-DETR-r18 against the full model. FPS was measured on an RTX 4090. No model uses pretrained weights.

Dataset and model	AP_{50}	$AP_{50:95}$	AP_S	FPS
SIMD, RT-DETR-r18	77.4	62.0	25.4	70.3
SIMD, HPS-DETR	79.8	63.5	27.9	80.7
DOTA tiles, RT-DETR-r18	85.1	60.7	37.0	68.8
DOTA tiles, HPS-DETR	86.8	62.5	39.2	80.2
NWPU VHR-10, RT-DETR-r18	88.4	57.5	26.5	74.3
NWPU VHR-10, HPS-DETR	91.5	60.4	29.8	82.8
<i>Strongest competitors on SIMD</i>				
YOLO11m	78.8	62.4		
DEIM	79.1	63.0		
RT-DETR-r50	78.3	62.7		

Its attention ablation is a template worth copying. Holding the rest of the network fixed, it swaps six attention modules on DOTA. Standard self-attention gives 84.8 AP_{50} at 74.5 FPS. Deformable attention gives 85.5 at 67.6. HiLo gives 86.0 at 76.6. Cascaded group attention gives 86.8 at 80.2. A state-space block can be added to such a table as one more row, so that it competes on equal terms.

LIMITATIONS.

- Its own loss table suggests that run-to-run variation is about 1 to 1.5 points. Inner-MPDIoU with ratio 1.0 is mathematically equivalent to MPDIoU, yet it scores 1.6 AP_{50} lower on SIMD (78.2 against 79.8). Several of the claimed gains are no larger than this.
- The small-object score AP_S on NWPU swings from 23.2 to 29.8 with a single constant in the upsampler.
- The DOTA experiments use the authors’ own tiles and an unstated box protocol, so they cannot be compared with the official leaderboard.
- Several citations do not point to the works the text attributes to them.
- The work uses visible light only.

WHAT TO TAKE FROM IT. The stride-4 level contributes most of the small-object gain. Global attention is affordable only at P_5 , which is precisely where a linear-time state-space block could be tried at P_3 and P_4 instead. Repeated runs are both possible and persuasive.

3.3 UAVDet: a CNN–Mamba hybrid for UAV imagery

WHAT IT PROPOSES. UAVDet, by Yang, Guo and Niu, is built on RTMDet-m in MMDetection, not on a YOLO code base [78, 156]. It makes three changes.

- The backbone is made heavier in its early stages and lighter in its late stages. Channels become 192–192–96–96 and block counts 4–2–2–1.
- A cross-stage partial Mamba block (CSPMB) is placed in the neck at P_3 and P_4 . It runs a VMamba-style four-direction scan beside a convolutional branch.
- The neck is redesigned to add a stride-4 head and remove the stride-32 head.

The paper links to no code.

EVIDENCE. Table 3.4 shows the comparison on the VisDrone test set and the ablation that explains it. The ablation is the most informative result in the four papers. On top of the other two changes, the Mamba block adds 0.7 AP while frame rate falls from 76 to 53. A transformer block in the same position is worse on both counts: 25.6 AP, 46 FPS and seven times the memory. The authors offer a Mamba-free variant for this reason. They also show that a plain path-aggregation neck with an added stride-4 head reaches 25.2 AP, the highest value in their neck study.

LIMITATIONS.

- The parameter counts are implausible. The full model is listed at 3.74M parameters, although its separate components are listed at 16.54M and 23.96M. The paper does not explain this, and a reimplementations should check it.
- AP_S for the baseline is 10.3 in three tables and 14.5 in two others.
- Frame rate is measured with an unstated protocol. The final model is slower than its baseline and slower than every YOLO in the comparison.
- All results come from single runs. A confidence threshold of 0.3 was used at inference, which can depress AP if the same threshold was applied during evaluation.
- The DroneVehicle experiments use the visible images only.

WHAT TO TAKE FROM IT. The most important lesson in the four papers comes from this ablation. On still images, a stride-4 head and early capacity buy several points. The state-space block buys less than one point and costs about a third of the frame rate.

TABLE 3.4. UAVDet results from the paper on the VisDrone2019 test set at 640×640 , measured on an RTX 4090. The upper block compares detectors and the lower block isolates each module. OB is the optimised backbone. TFPN is the redesigned neck with a stride-4 head.

Model	AP	AP ₅₀	AP _S	Params (M)	GFLOPs	FPS
YOLOv8m	19.3	33.9	9.4	25.9	79.1	106
YOLO11m	20.6	35.7	10.7	20.1	68.2	133
Mamba YOLO-M	18.3	32.2	9.0	21.8	49.6	141
RemDet-m	23.3	39.2	11.6	23.3	34.4	188
RT-DETR	24.5	42.3	13.5	42	136	68
RTMDet-m (baseline)	22.3	37.5	10.3	24.71	39.27	59
UAVDet	26.8	44.8	15.3	3.74 [†]	46.63	53
<i>Ablation, each row relative to the baseline</i>						
+ OB	25.6	43.5	14.3	23.96	42.68	
+ CSPMB only	22.9	38.3	10.8	29.23	43.80	
+ TFPN only	24.7	42.0	13.5	16.54	39.03	
+ OB, TFPN	26.1	44.0	15.2	3.40 [†]	45.18	76
+ OB, TFPN, transformer	25.6	42.9		3.54	54.20	46
+ OB, TFPN, CSPMB	26.8	44.8	15.3	3.74 [†]	46.63	53

3.4 CFMW: cross-modality fusion Mamba under adverse weather

WHAT IT PROPOSES. CFMW, by Li and colleagues, targets visible–thermal detection in rain, fog and snow [69]. It has two parts. A conditional diffusion model first restores the weather-degraded visible image. It follows WeatherDiff and uses DDIM sampling [55, 98, 128]. A dual-stream YOLOv5-L then extracts features from both images and fuses them at three backbone stages with a cross-modality fusion Mamba block. That block does four things in turn.

- It swaps half of the channels between the two streams.
- It normalises and projects each stream.
- It applies VMamba’s four-direction scan to each stream.
- It gates the outputs and adds them back to both streams through crossed residuals.

The paper also releases SWVI, a benchmark of 64 281 paired images with synthetic rain, fog and snow, built from LLVIP, M3FD, MSRS and FLIR.

EVIDENCE. Table 3.5 gives the main comparison and the efficiency study. The efficiency figures are the most revealing. The state-space block triples computation, from 95.6 to 308.6 GFLOPs, for 3.2 points of mAP. The full model runs at 13.5 frames per second, which is not

real-time. On LLVIP, the thermal-only YOLOv10 scores 62.5 mAP. The plain two-stream YOLOv10 scores 63.4. Fusion therefore adds less than one point on that dataset before any new module is introduced.

TABLE 3.5. CFMW results read from Tables IV, V and VII of the paper. mAP is averaged over IoU 0.5 to 0.95. The two values marked with a dagger disagree with other parts of the paper (Table 3.7).

Model, input	mAP ₅₀	mAP ₇₅	mAP	GFLOPs	FPS
<i>LLVIP</i>					
YOLOv10, thermal only	95.3	72.4	62.5		
YOLOv10, two-stream	96.1	72.7	63.4		
CFT, two-stream	97.5	72.9	63.6		
CFMW	98.8	77.2	69.8 [†]		
<i>SWVI</i>					
YOLOv10, thermal only	94.7	70.9	62.7		
YOLOv10, two-stream	92.1	68.2	59.3		
CFT, two-stream	94.4	69.7	59.4		
CFMW	97.2	75.9	68.4 [†]		
<i>SWVI efficiency study at 640 × 640</i>					
CFT	93.4	71.8	59.7	290.73	5.14
CFMW without channel swap and scan	92.2	70.6	58.4	95.58	23.64
CFMW without the scan	94.8	72.3	60.2	95.58	19.54
CFMW without channel swap	96.4	75.8	62.5	308.62	15.73
CFMW, full	97.2	76.9	63.4	308.62	13.52

LIMITATIONS.

- The weather is synthetic, and only the visible image is degraded. Real fog and rain also affect thermal sensors.
- The method depends on pixel-aligned pairs. Channel swapping and element-wise addition assume that the two images correspond pixel for pixel.
- Training the restoration model takes about three days on a 48 GB A6000.
- Epochs and input size for the detector are not stated in the text.
- The authors note that small objects remain a failure case. Objects below 2 500 square pixels make up 26.2% of SWVI.
- Fusion-Mamba, ICAFusion and other contemporaries are not compared.

WHAT TO TAKE FROM IT. CFMW is already the journal version of the static design implied by the project title. It shows how the design is built and evaluated. It also shows the price: two backbones, a scan at three stages and a frame rate far below real time. Any new contribution must either avoid that price or buy something with it that still images cannot.

3.5 What the four papers show together

3.5.1 Where the gains come from

Figure 3.1 collects every isolated measurement in the four papers, and in two closely related works, of what the state-space component contributes. These are set beside the contributions of other components in the same papers. The picture is consistent. On still images, the state-space part adds between -1.3 and $+3.2$ points, usually below one point and often at a large cost in speed or computation. The stride-4 head and redistributed early capacity add more, and they do so cheaply.

3.5.2 What none of them do

Table 3.6 lists the capabilities a journal reviewer would look for. None of the four papers processes video, carries memory across frames, tests robustness to sensor failure or measures latency on an edge device. Only one repeats its experiments. These gaps define where a new contribution can stand.

TABLE 3.6. Capabilities covered by the four source papers. Entries are as reported in each paper.

Capability	MambaVision	HPS-DETR	UAVDet	CFMW
state-space component	yes	no	yes, neck	yes, fusion
thermal input	no	no	no	yes
small-object metric	no	AP_S	AP_S, AP_T	size analysis
video or temporal memory	no	no	no	no
robustness to sensor faults	no	no	no	weather only
repeated runs	no	four	no	no
latency at batch one	no	yes, 4090	yes, 4090	yes, GPU unstated
edge deployment	no	no	no	no
code released	yes	no	no	announced

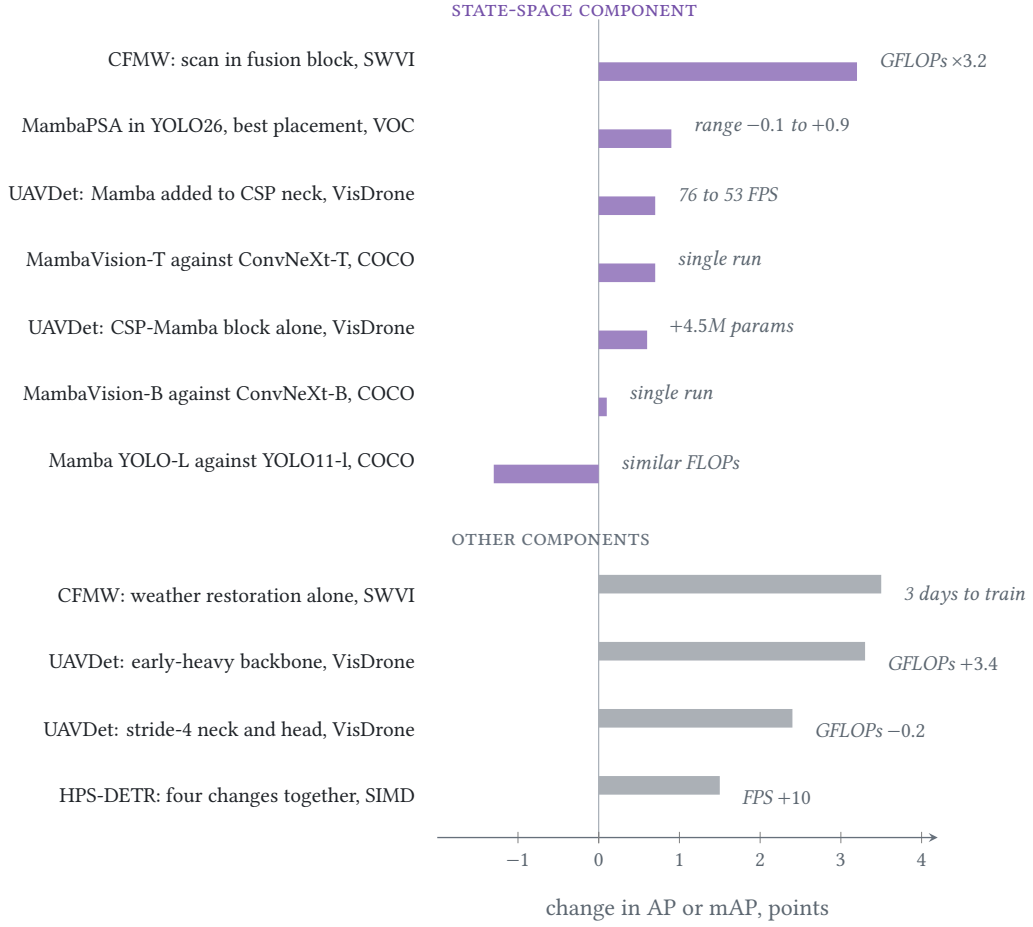


FIGURE 3.1. Measured contribution of the state-space component (purple) against other components (grey), from the ablations of the source papers and two related works [51, 69, 82, 145, 146, 156, 158]. Each bar isolates one change on one dataset. Notes on the right give the cost reported with it. On still images, the stride-4 pathway and the redistribution of early capacity contribute more than the state-space block, and at lower cost.

3.5.3 Reporting problems

All four papers contain internal inconsistencies (Table 3.7). None of them changes the main conclusions. Together they argue for two habits. Every baseline in the new work should be re-run under one protocol, not copied from papers. Every table should be generated directly from logged runs (Section 10.5).

TABLE 3.7. Inconsistencies found in the source papers. Values are quoted as printed.

Paper	Inconsistency
MambaVision	Ablation text gives a baseline of 80.9% top-1 and 40.2 mask AP; the table gives 80.5% and 40.4. Claims 56% fewer GFLOPs than MaxViT-B; 15.0 against 23.4 is 36% fewer. The output projection in Eq. 7 has the wrong input dimension.
HPS-DETR	Inner-MPDIoU at ratio 1.0, equivalent to MPDIoU, differs from it by 1.6 AP_{50}. The new backbone block is said to speed the model up, yet DOTA FPS falls from 68.8 to 65.2. References [4], [20], [30] and [34] do not match the works described.
UAVDet	The full model has 3.74M parameters although its components have 16.54M and 23.96M. Baseline AP_S is 10.3 in Tables 3, 5 and 6 and 14.5 in Tables 7 and 8. CSPMB at L_3-L_5 has 29.23M parameters in Table 5 and 29.41M in Table 6.
CFMW	LLVIP mAP is 69.8 in Table IV and 64.8 in the text. The full model scores 68.4 mAP on SWVI in Table V and 63.4 in Tables VI and VII. The SWVI splits sum to 59 990 images, not the stated 64 281. A 51.2% memory saving is claimed; 18.0 against 35.2 GB is 48.9%. Three fusion blocks are described, but performance peaks at eight in Fig. 7.

State of the Art, October 2026

This chapter surveys the six areas the project draws on. In each, the newest verified work comes first. The search closed on 2 October 2026. Each table gives the month a work first appeared, its venue where one exists, and the result most relevant here. Older work appears only where it is still the reference baseline. The aim is not an exhaustive review but a reliable map of the frontier that the proposed method must be measured against.

4.1 Real-time general detectors

The accuracy–latency frontier moved in two ways between 2024 and 2026. The YOLO line kept improving its training recipe and heads. YOLOv10 removed the need for NMS [144]. YOLOv12 made attention efficient enough for real time [132]. YOLOv13 added hypergraph correlation modules [65]. YOLO26 removed distribution focal loss and added a small-target-aware label assignment [160].

At the same time, detection transformers began to draw on large self-supervised vision models. RF-DETR uses a DINOv2 backbone with weight-sharing architecture search [97, 111]. DEIMv2 and RT-DETRv4 use DINOv3, as features or as a teacher during training [23, 113, 124]. RF-DETR is the first real-time detector reported above 60 COCO AP. Table 4.1 and Figure 4.1 summarise the field.

For this project, YOLO26 is the natural base. It is the newest YOLO and is maintained in the Ultralytics library [160]. Its label assignment already targets small objects. Its modules are defined in a configuration file, which makes new blocks easy to register. The DETR frontier is stronger at equal latency, but it is heavier to train on a T4 GPU. It is better treated as a reference point than as a base.

TABLE 4.1. Real-time detectors, newest first. COCO val2017 AP and latency on a T4 GPU with TensorRT FP16, as reported by each work. The ranges run from the smallest to the largest model.

Date	Model	Venue	COCO AP and latency
06/2026	YOLO26 [160]	arXiv; released 01/2026	40.9 to 57.5 at 1.7 to 11.8 ms; optional NMS-free head; small-target-aware assignment
11/2025	RF-DETR [111]	ICLR 2026	48.4 to 60.1 at 2.3 to 17.2 ms; DINOv2 backbone
10/2025	RT-DETRv4 [113]	ECCV 2026	49.8 to 57.0 at 3.7 to 12.9 ms; DINOv3 teacher, no inference cost
09/2025	DEIMv2 [23]	arXiv	38.5 (1.5M params) to 57.8; first model under 10M params above 50 AP
06/2025	YOLOv13 [65]	arXiv	41.6 to 54.8 at 2.0 to 14.7 ms
02/2025	YOLOv12 [132]	NeurIPS 2025	40.4 to 55.4 at 1.6 to 10.4 ms; needs FlashAttention for full speed
12/2024	DEIM [58]	CVPR 2025	43.0 to 56.5; dense one-to-one matching, half the training time
10/2024	D-FINE [100]	ICLR 2025	42.8 to 55.8; 59.3 with Objects365 pretraining
09/2024	YOLO11 [158]	software	39.5 to 54.7 at 1.5 to 11.3 ms
05/2024	YOLOv10 [144]	NeurIPS 2024	38.5 to 54.4 at 1.8 to 10.7 ms; first NMS-free YOLO
04/2023	RT-DETR [169]	CVPR 2024	baseline of HPS-DETR; v2 and v3 followed in 2024
<i>Accuracy ceilings, not real-time</i>			
08/2025	DINOv3 7B with Plain-DETR [124]	arXiv	65.6 AP [†]
11/2022	Co-DETR, ViT-L [173]	ICCV 2023	66.0 AP on test-dev

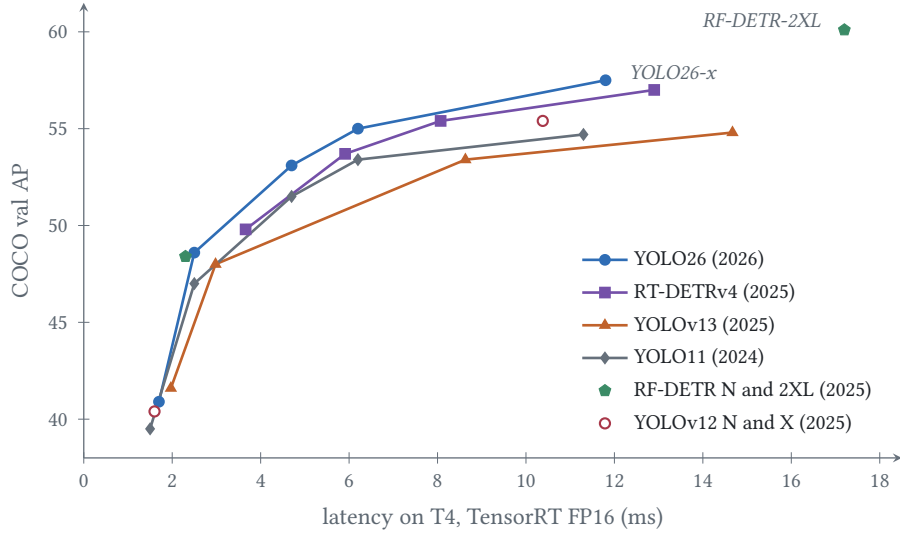


FIGURE 4.1. Accuracy against latency for current real-time detectors, using the figures each work reports [65, 111, 113, 132, 158, 160]. Lines join model sizes whose individual latencies were confirmed. For RF-DETR and YOLOv12, only the smallest and largest models are plotted. Protocols differ, notably in whether NMS is timed, so small horizontal differences should not be read as significant.

4.2 Vision state-space models and Mamba detectors

Vision state-space models are now competitive with Swin and ConvNeXt as general backbones. Examples include VMamba, Spatial-Mamba, DAMamba, A2Mamba and PRIS-Mamba [1, 17, 76, 102, 151]. Inside real-time YOLO detectors the evidence is weaker. Mamba YOLO, the AAI 2025 baseline that most later work builds on, reaches 52.1 COCO AP with its large model. YOLO11-l reaches 53.4 at similar computation [146, 158]. The closest work to the project title is MambaPSA (July 2026). It replaces YOLO26’s attention block with a Mamba block and changes AP by -0.1 to $+0.9$ on VOC [82]. Research effort in 2026 has moved to the scan order itself (Table 4.2).

4.3 Small and tiny objects

Small-object results are hard to compare because three choices vary between papers: the input resolution (640, 960 or about 1333 pixels), the split (validation or test-dev) and the metric (AP_{50} or COCO AP). Published VisDrone figures range from 26 to 39 AP, and up to 60 AP_{50} at high resolution, largely for these reasons. Table 4.3 reports each result with its protocol where the paper states it.

TABLE 4.2. Vision state-space models and Mamba detectors, newest first.

Date	Work	Venue	Contribution and result
09/2026	YOLO12-MambaScan [159]	arXiv	YOLO12 with Mamba context; VisDrone at 960 px, 38.6 AP
09/2026	ScopeMamba-YOLO [117]	arXiv	off-path selective scan; VisDrone 50.8 AP ₅₀ with 3.6M params
07/2026	MambaPSA [82]	arXiv	Mamba in place of YOLO26’s C2PSA; 12% fewer FLOPs, −0.1 to +0.9 AP
07/2026	VNCT [140]	arXiv	non-causal trapezoidal SSM; argues directional scans are unnecessary
06/2026	AKCMamba-YOLO [3]	CVPR 2026	channel-interaction Mamba modules in YOLO
05/2026	C-GSPN [9]	arXiv	notes that SSMs weaken 2D structure; 4× faster at 2K resolution
03/2026	SF-Mamba [120]	arXiv	reports Mamba is slow at short token lengths; batch folding as remedy
02/2026	PRISMamba [102]	ICML 2026	partial ring scan; 84.5% ImageNet top-1 at 3.9 GFLOPs
07/2025	A2Mamba [1]	arXiv	+1.2 box AP over MambaVision-B with 40% fewer parameters
06/2025	MambaNeXt-YOLO [81]	arXiv	CNN–Mamba block; runs on Jetson Xavier and Orin NX
02/2025	DAMamba [17]	NeurIPS 2025	dynamic adaptive scan; 48.5 to 50.6 box AP, Mask R-CNN 1×
10/2024	Spatial-Mamba [151]	ICLR 2025	structure-aware state fusion; 47.6 to 50.4 box AP
07/2024	MambaVision [51]	CVPR 2025	source paper; Section 3.1
06/2024	Mamba YOLO [146]	AAAI 2025	Ultralytics-based baseline; COCO 44.5 to 52.1 AP
05/2024	MambaOut [163]	CVPR 2025	SSM unnecessary for classification; may help detection
05/2024	MLLA [50]	NeurIPS 2024	Mamba as gated linear attention; forget gate matters most
01/2024	VMamba [76]	NeurIPS 2024	four-direction SS2D scan
01/2024	Vim [172]	ICML 2024	bidirectional scan

TABLE 4.3. Small and tiny object detection, newest first.

Date	Method	Venue	Reported result
06/2026	SFDNet [119]	ECCV 2026	spectrum-aware; SODA-D 34.2 AP, SODA-A 39.2 AP, AI-TOD 31.7 AP
06/2026	DERNet [25]	arXiv	frequency domain; beats YOLO11 with one sixth of the parameters
03/2026	O2-DEIM [95]	IEEE TGRS	oriented real-time DETR; DOTA-v1.0 80.15 AP ₅₀ at 119 FPS
01/2026	EFSI-DETR [34]	arXiv	VisDrone 33.1 AP, 24.8 AP _S [†]
01/2026	TOLF [135]	arXiv	normalising flows; +1.2 AP over DINO on AI-TOD
05/2025	Dome-DETR [28]	ACM MM 2025	AI-TOD-V2 34.7 AP; VisDrone 39.0 AP at about 376 GFLOPs
01/2025	Strip R-CNN [164]	AAAI 2026	DOTA-v1.0 82.75 mAP with 30M params
01/2025	UAV-DETR [138]	arXiv	+3.1 AP over RT-DETR on VisDrone
04/2024	DQ-DETR [56]	ECCV 2024	AI-TOD-V2 30.2 AP
2021	NWD [143]	ISPRS J. 2022	Wasserstein matching for tiny boxes

4.4 Visible–thermal fusion

Fusion has followed the same arc as general detection. Attention-based fusion came first: CFT and ICAFusion [105, 122]. State-space fusion followed in 2024 and 2025. Its main members are Fusion-Mamba, COMO, WaveMamba and MS2Fusion [14, 29, 91, 148]. In 2026, methods built on foundation models appeared. RegisterBridgeMM, for example, uses a frozen DINOv3 encoder [107]. Table 4.4 shows the four standard benchmarks.

LLVIP is saturated. Every recent method scores 97 to 98.4 AP₅₀, and a thermal-only model is close behind. DroneVehicle results are not comparable across papers, because some use horizontal and some use oriented boxes. M3FD has no official detection split. KAIST is scored by miss rate, not AP, and has no clear 2025–2026 leader.

4.5 Video, tracking and foundation models

The standard tracking benchmarks MOT17 and MOT20 have levelled off at about 64 to 66 HOTA. New work reports mainly on DanceTrack and SportsMOT, which test association under uniform appearance and fast motion (Table 4.5). SambaMOTR is the only state-space tracker among the leaders. It keeps a synchronised SSM memory per track [118].

TABLE 4.4. Visible–thermal fusion detectors, newest first. Each cell gives AP_{50} and $mAP_{50:95}$. A dash means not reported.

Date	Method	LLVIP	M3FD	FLIR-aligned	DroneVehicle
08/2026	RegisterBridgeMM [107]	98.4 / 70.5	92.6 / 64.9	88.9 / 49.8	81.5 / 61.5
08/2026	CFGPNNet [8]	97.8 / 68.9	89.9 / 63.4	80.7 / 45.0	–
11/2025	MambaRefine-YOLO [84]	–	–	–	83.2 AP_{50}
07/2025	WaveMamba [148]	98.3 / 66.0	92.1 / 64.4	88.4 / 48.1	79.8 / 60.5
07/2025	MS2Fusion [91]	reported	reported	reported	–
12/2024	COMO [14]	97.2 / 65.3	–	–	86.1 / 65.5
04/2024	Fusion-Mamba [29]	97.0 / 64.3	88.0 / 61.9	84.9 / 47.0	–
04/2024	CFMW [69]	98.8 / 69.8 [†]	–	–	–
08/2023	ICAFusion [122]	–	81.5 / 54.5	–	–
11/2021	CFT [105]	97.5 / 63.6	–	–	84.3 / 61.9

For video object detection on ImageNet VID, no strong 2025–2026 method was found besides TMambaDet [134]. TMambaDet uses a bidirectional, and therefore offline, temporal Mamba.

Foundation models now reach the detection task directly. SAM 3 detects, segments and tracks objects named by a text prompt [115]. YOLOE brings open-vocabulary prompts to the YOLO architecture [147]. DINOv3 has become the backbone or teacher of choice for detection transformers [124].

TABLE 4.5. Multi-object tracking, newest first. HOTA on DanceTrack and SportsMOT test sets as reported.

Date	Method	Venue	DanceTrack	SportsMOT	Note
07/2026	SpikingMOT [129]	arXiv	56.5	74.9	spiking network
06/2026	SAM 3 mask propagation [127]	arXiv	–	87.2 [†]	offline, training-free
09/2025	MATR [88]	arXiv	71.3	72.7	with extra data
03/2024	MOTIP [41]	CVPR 2025	67.5	71.9	tracking as ID prediction
10/2024	SambaMOTR [118]	ICLR 2025	67.2	69.8	state-space track memory
11/2022	MOTRv2 [168]	CVPR 2023	69.9 [†]	–	YOLOX proposals
03/2022	OC-SORT [6]	CVPR 2023	55.1 [†]	68.1	motion-based
10/2021	ByteTrack [166]	ECCV 2022	47.7 [†]	–	MOT17 HOTA 63.1

4.6 Caveats that govern fair comparison

Seven problems recur across the literature. The experimental protocol in Chapter 8 is designed around them.

1. *Mamba is not automatically fast.* Its scan beats FlashAttention-2 only beyond about 2 000 tokens [18, 48]. YOLO’s P_4 and P_5 maps are shorter than that.
2. *Installation.* The official `mamba_ssm` kernels support Linux only [85]. On Windows, the choices are WSL2, a cloud notebook, or a pure-PyTorch scan that is about twice as slow [83].
3. *Scan order.* A one-dimensional causal scan breaks two-dimensional adjacency. At least five papers in 2026 propose remedies, so a fixed order needs justification.
4. *Baseline drift.* Published numbers for the same baseline differ between papers. YOLOv12-N, for example, appears as 40.6, 40.4 and 40.1 AP. Ultralytics itself recommends YOLO11 or YOLO26 over YOLO12 for stability [160].
5. *Latency protocol.* Many papers omit NMS from latency. GPU thermal throttling can shift measurements by about 25%. Naive FP16 export can collapse accuracy. RF-DETR’s protocol addresses all three: it includes NMS, pauses between passes and measures accuracy on the exported engine [111].
6. *Saturated benchmarks.* COCO contains label errors that COCO-ReM corrects [125]. LLVIP is effectively solved. RF100-VL shows that COCO gains often fail to transfer to new domains [110].
7. *Metric choice.* Small-object papers often report AP_{50} at high resolution. AP, AP_{50} and size-specific AP should all be reported, at a fixed resolution and against baselines trained with the same schedule.

4.7 Implications for this project

The survey settles four choices.

- The base detector should be YOLO26. A DETR reference point should be kept for honesty.
- State-space blocks should be placed where they are efficient and where small objects live, at P_3 and below, or along the time axis, where sequences are long.
- Fusion results should be reported on benchmarks that are not saturated, with $mAP_{50:95}$.

- The decisive comparison is against temporal and fusion baselines from 2025 and 2026, not against YOLOv8.

Novelty Gap Analysis

A journal contribution has to be new against the work that exists on the day of submission, not on the day the project was proposed. This chapter maps 64 papers from 2024 to September 2026 that are close to the project. It identifies which directions are saturated and states precisely the one that remains open.

5.1 A saturated design space

Figure 5.1 places the papers on a timeline in four lanes.

- The first lane, *Mamba inside YOLO or for small objects*, holds 24 papers. Three of them appeared in September 2026 alone: ScopeMamba-YOLO, YOLO12-MambaScan and HDMamba-YOLO [52, 117, 159]. There is also a CVPR 2026 paper [3] and a steady flow of journal articles.
- The second lane, *Mamba for visible–thermal fusion*, holds 17 papers. They appeared in ICCV, IEEE TMM, IEEE TCSVT and two volumes of Information Fusion.
- The third lane covers state-space models used *across time* for any task. It is dominated by tracking and by event cameras.
- The fourth lane covers *visible–thermal video detection* without state-space models. It uses windows, graphs and hypergraphs.

The design implied by a literal reading of the project title is a YOLO detector with Mamba blocks fusing visible and thermal features on still images. It sits at the intersection of the first two lanes. Fusion-Mamba describes itself as “a dual-stream feature extraction network and three Fusion-Mamba blocks, with the same neck and head as YOLOv8” [29]. MambaRefine-

YOLO, RemoteDet-Mamba, WaveMamba and COMO occupy the same point [14, 84, 108, 148]. A reviewer at IEEE TCSVT or Pattern Recognition would read a further instance as incremental.

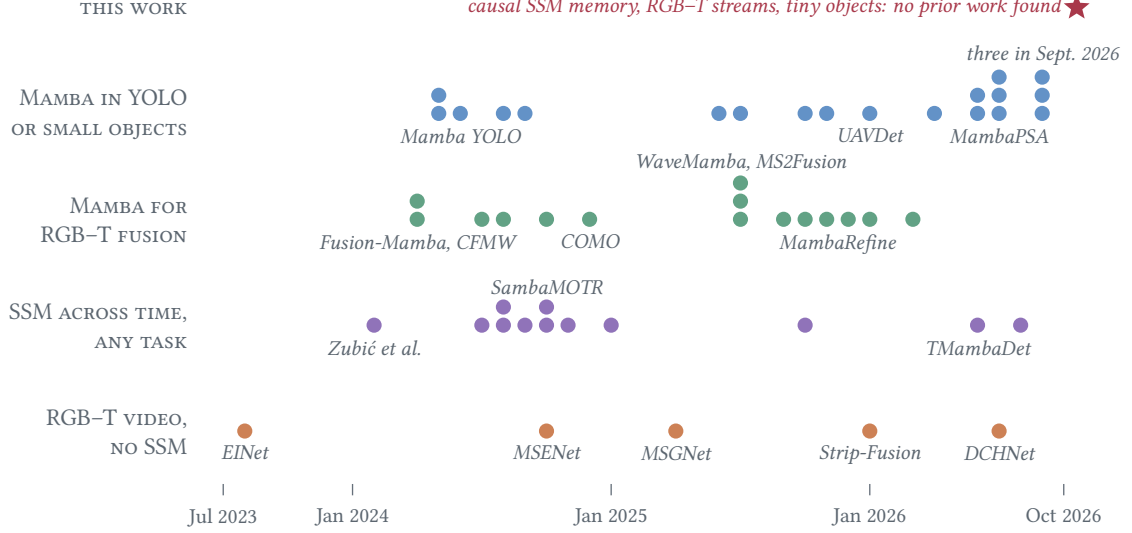


FIGURE 5.1. Timeline of closely related work from July 2023 to October 2026, one dot per paper, stacked when several share a month. Papers without a known month of publication are omitted. Static Mamba-in-YOLO and Mamba fusion papers accumulate quickly. Temporal state-space work concentrates on tracking and event cameras. Visible-thermal video detection has advanced without state-space models. No work occupies the top lane.

5.2 Five candidate directions

Table 5.1 assesses five directions that would extend the static design. Each is judged by the closest prior art found and by what remains open.

Direction (c) is crowded and should be avoided as a main claim. Directions (d) and (e) are valuable but either too large for one paper or too narrow for a top journal. Directions (a) and (b) are complementary. Combined, they define a problem no published work addresses.

5.3 The open problem

Stated precisely, the open problem is this:

Detect very small objects in a continuous stream of visible and thermal frames, using only past frames, at constant memory and per-frame cost, while remaining

TABLE 5.1. Candidate novelty directions and their prior art. Risk is the likelihood that a reviewer finds the contribution already published.

Direction	Closest prior art	What remains open	Risk
(a) causal state carried across frames	MambaST [86]; Zubić et al. [174]; TMambaDet [134]; StreamYOLO [155]	No causal, constant-memory, unbounded-length visible–thermal detector. No state-space method on RGB–thermal video detection benchmarks.	low to medium
(b) misalignment and missing modality	COMO [14]; contextually guided SSM fusion [10]; MSDF-Mamba [92]; SCDT for tracking [116]	Temporal missing modality in detection, such as bursty thermal dropout and desynchronisation, has no benchmark protocol.	medium; low if combined with (a)
(c) size-aware scan order	QuadMamba [152]; DAMamba [17]; ObjM [96]; Scope-Mamba [117]	Only a temporal twist, ordering tokens by objects predicted in the previous frame.	high
(d) joint detection and tracking with shared state	SambaMOTR [118]; MambaTrack [87]; TrackSSM [137]	A shared state for dense YOLO-style detection and identity in RGB–thermal tiny-object tracking.	medium; large scope
(e) edge deployment of SSM detectors	MambaNeXt-YOLO [81]; LCMamNet [64]; DLR-Mamba [27]	No native TensorRT operator for two-dimensional selective scan was found; accuracy loss after export is rarely reported.	medium to high alone

accurate when one sensor drops out, drifts out of alignment or delivers frames at irregular intervals.

Figure 5.2 places the closest work on two axes. The first is how much temporal context a method uses. The second is how demanding its sensing assumptions are. Every cell to the left of the last column is occupied. In the last column, causal and unbounded memory exists only for event cameras and for DETR-style tracking. The two cells in the lower right are empty. Those cells are the target of CFFM-Net-S.

		<i>temporal context used</i>		
		SINGLE IMAGE	SHORT WINDOW OF FRAMES	CAUSAL MEMORY, UNBOUNDED LENGTH
<i>sensing assumptions, increasingly demanding</i>	VISIBLE ONLY	Mamba YOLO, MambaPSA, UAVDet, HPS-DETR, MambaVision, YOLO26	TMambaDet (offline), StreamYOLO, LongShortNet, DAMO-StreamNet	Zubić et al., RVT (event cameras); SambaMOTR (memory per track)
	VISIBLE AND THERMAL, ALIGNED	Fusion-Mamba, CFMW, WaveMamba, MS2Fusion, MambaRefine-YOLO	MambaST (3 frames, KAIST only), EInet, Strip-Fusion	empty target of CFFM-Net-S: temporal state memory
	VISIBLE AND THERMAL, MISALIGNED OR MISSING FRAMES	COMO, contextually guided SSM fusion, MSDF-Mamba, DMM	MSENet, MSGNet, MCDA-Net, DCHNet (alignment-free)	empty target of CFFM-Net-S: reliability-gated Δ

FIGURE 5.2. Positioning of the closest work. Columns give the temporal context a method uses. Rows give how demanding its sensing assumptions are. Each cell lists representative methods. The two highlighted cells contain no published method that the searches could find. They are the targets of CFFM-Net-S.

5.4 Positioning against the nearest work

Six papers come closest, and the contribution must be defined against each (Table 5.2).

- MambaST is the nearest in spirit [86]. It fuses visible and thermal features across three frames with a stride of three. It feeds the last hidden output into the next window and is evaluated on KAIST only. Its authors name longer sequences and other datasets as future work.

- Zubić and colleagues show truly recurrent state-space detection with frequency generalisation, on event data [174].
- TMambaDet is bidirectional and therefore offline [134].
- SambaMOTR keeps state per track, not per location. It is built on a DETR tracker [118].
- COMO handles misalignment on still images only [14].
- RVT is a recurrent detector for event cameras built on convolutional LSTM cells, not state-space models [43].

TABLE 5.2. Properties of the nearest prior work against the proposed CFFM-Net-S.

Property	MambaST	Zubić	TMamba-Det	Samba-MOTR	COMO	CFFM-Net-S
detection of objects	yes	yes	yes	tracks	yes	yes
uses only past frames	yes	yes	no	yes	n/a	yes
memory unbounded in length	no	yes	no	yes	n/a	yes
constant per-frame cost	no	yes	no	yes	n/a	yes
visible and thermal input	yes	no	no	no	yes	yes
small-object benchmark	no	no	no	no	partly	yes
robust to misalignment	no	n/a	n/a	n/a	yes	yes
robust to sensor dropout	no	n/a	n/a	n/a	no	yes
handles irregular frame rate	no	yes	no	no	n/a	yes

5.5 What the contribution is not

Three claims will be avoided.

- The work will not claim a new static fusion block as its contribution. The still-image CFFM-Net is a baseline and an ablation, not the headline.
- It will not claim a new scan order.
- It will not claim state-of-the-art accuracy on COCO or on saturated fusion benchmarks.

The claim is narrower and stronger: a causal, constant-memory, reliability-aware state-space memory for visible–thermal streams, together with a robustness protocol that tests it. The searches behind this chapter found no such method. That is evidence, not proof, and the search must be repeated immediately before submission (Section 12.6).

The Proposed Method: CFFM-Net

This chapter specifies CFFM-Net, the Cross-Feature Fusion Mamba Network, and its streaming variant CFFM-Net-S. The still-image network fulfils the supervised topic directly. It is a YOLO detector whose visible and thermal features are fused by state-space blocks. The streaming variant adds the contribution identified in Chapter 5: a causal state carried from frame to frame. Every component has an ablation in Chapter 8, and every claim is stated as a hypothesis in Section 6.9.

6.1 Design principles

Five principles follow from the evidence of Chapters 3 and 4.

- p1. *Keep what is known to work.* The base is YOLO26, with its small-target-aware assignment. A stride-4 head and stronger early stages are added, because those components produced the largest and cheapest gains in the source papers (Figure 3.1).
- p2. *Spend state-space computation where sequences are long.* Within a frame, scans are applied at P_3 to P_5 , where interleaving the two modalities doubles the sequence length. Across frames, the sequence is the stream itself and has no fixed length.
- p3. *Use time.* Memory carried across frames is the capability that none of the source papers, and none of the static fusion papers, provide.
- p4. *One control for three problems.* The discretisation step Δ is modulated by sensor reliability, by the true frame interval and by scene changes. In each case it decides how much the state should hold and how much it should write (Section 2.3.1).

- p5. *Deployable by construction.* Every block has a pure-PyTorch path and a single-step form that updates the state once per frame, so that the streaming model can be exported and run on live cameras.

6.2 Architecture overview

Figure 6.1 shows the full network for one frame t . Visible and thermal frames enter two YOLO26 backbones. The backbones have separate weights, because the two sensors have different statistics; sharing the later stages is an ablation. Each produces features $F_t^{v,l}$ and $F_t^{\theta,l}$ at levels $l \in \{2, 3, 4, 5\}$. Here v denotes visible and θ thermal.

At P_2 , fusion is a reliability-weighted sum, which is cheap enough for 160×160 maps. At P_3 to P_5 , the Cross-Modal Fusion Mamba (CMFM) block of Section 6.4 produces fused features Z_t^l and reliability maps ρ_t^l . In the streaming variant, the Temporal State Memory (TSM) of Section 6.5 combines Z_t^l with the state carried from frame $t - 1$. Before use, that state is warped by the estimated camera motion $H_{t-1 \rightarrow t}$. The fused, temporally enriched features enter a YOLO26 path-aggregation neck extended to P_2 . Heads at P_2 , P_3 and P_4 follow, with the P_5 head optional. Two side modules supply the signals that control Δ .

- The *stream monitor* reports the true interval τ_t since the last processed frame, sensor availability flags and a scene-cut test.
- The *ego-motion estimator* computes $H_{t-1 \rightarrow t}$.

6.3 The small-object pathway

Four measures protect small objects before any state-space block is involved.

- *Stride-4 head.* The P_2 head is added and the neck carries P_2 through both its paths. In UAVDet and HPS-DETR this was the largest single source of small-object gain.
- *Early capacity.* Channels and blocks are moved towards the first two stages, following UAVDet’s ablation. The backbone configuration is a hyperparameter with three settings in the ablation.
- *Native resolution.* RGBT-Tiny frames are 640×512 and are processed at native resolution with letterbox padding, so no downscaling destroys tiny objects. For larger inputs, sliced inference with SAHI [4] is an inference option, not a training change.

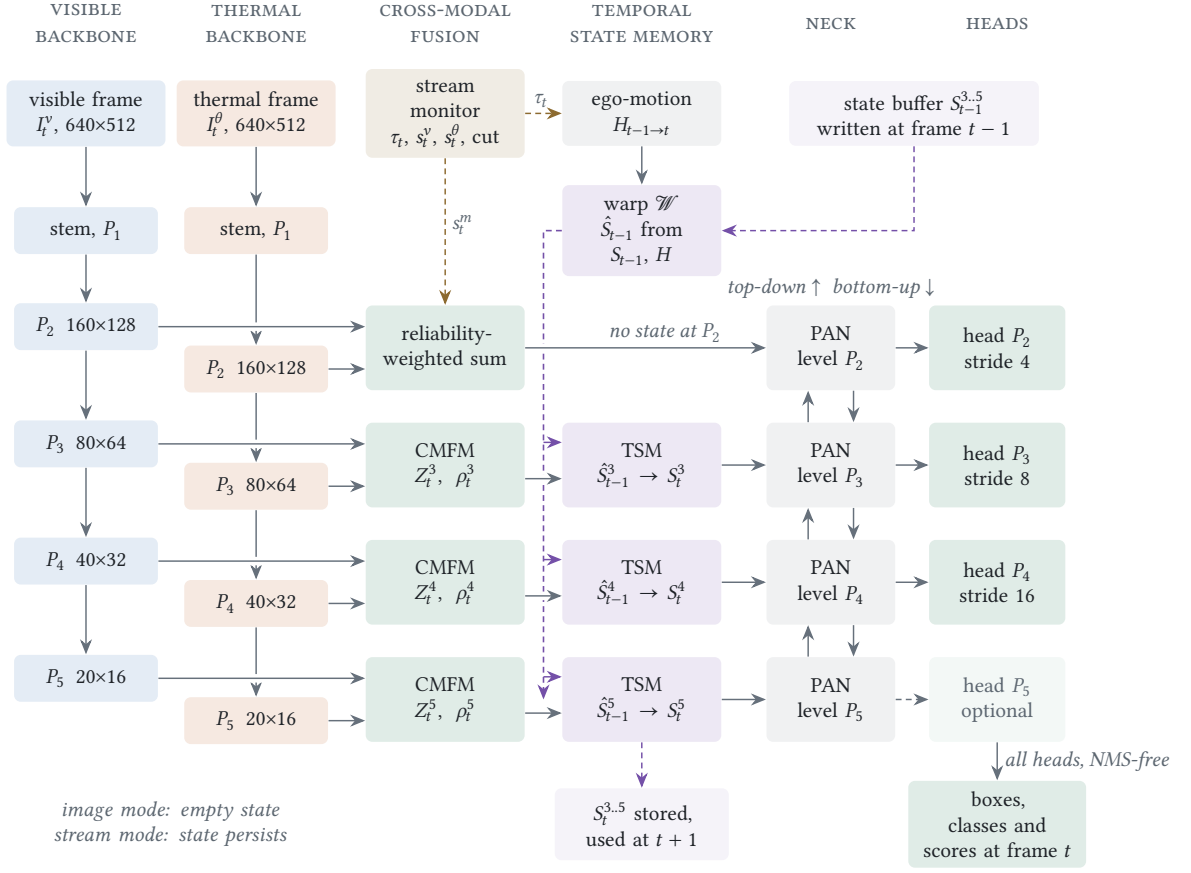


FIGURE 6.1. Architecture of CFFM-Net and CFFM-Net-S for one frame of a 640×512 visible-thermal stream. Two YOLO26 backbones produce features at strides 4 to 32. A reliability-weighted sum fuses P_2 . CMFM blocks fuse P_3 to P_5 and emit reliability maps ρ . In stream mode, the Temporal State Memory updates a per-location state carried from frame $t - 1$ after warping by the estimated camera motion. Purple dashed paths carry state. Ochre dashed paths carry the monitor’s signals: sensor flags s_t^m , interval τ_t and scene cuts. The neck and heads follow YOLO26 with a stride-4 extension. In image mode the state is empty and the network reduces to the still-image CFFM-Net.

- *Assignment.* YOLO26’s small-target-aware label assignment is retained [160]. The scale-adaptive SAFit measure proposed with RGBT-Tiny is used in evaluation [157].

6.4 Cross-Modal Fusion Mamba with reliability-gated steps

The CMFM block fuses $F^v, F^\theta \in \mathbb{R}^{C \times H \times W}$ at one pyramid level. Superscript l and subscript t are dropped for clarity. Figure 6.2 shows its structure. It has five steps.

RELIABILITY ESTIMATION. A small convolutional head g estimates, at every location, how far each modality can be trusted:

$$(r^v, r^\theta) = \sigma\left(g\left(\begin{bmatrix} F^v & F^\theta & |F^v - F^\theta| \end{bmatrix}\right)\right) \odot (s^v, s^\theta), \quad r^v, r^\theta \in (0, 1)^{H \times W}. \quad (6.1)$$

Here g is a depthwise 3×3 convolution followed by a 1×1 convolution with two outputs. The flags $s^v, s^\theta \in \{0, 1\}$ come from the stream monitor. They are zero when a sensor delivers no frame, so that a known outage needs no learning. The absolute difference is included because disagreement between modalities is the most direct evidence that one of them is degraded.

OFFSET-GUIDED ALIGNMENT. A second head predicts a displacement field $o \in \mathbb{R}^{2 \times H \times W}$ from $[F^v, F^\theta]$. The thermal features are resampled onto the visible grid, $\tilde{F}^\theta = \mathcal{S}(F^\theta, o)$, with bilinear sampling. This follows the offset guidance of COMO [14]. The field is initialised to zero, so the block starts as an aligned fusion. Its effect is measured under synthetic misalignment (Section 8.4).

TOKEN INTERLEAVING. Both streams are projected to C' channels by 1×1 convolutions ϕ_v and ϕ_θ . They are flattened along the scan order and interleaved, so that the two modalities at each location are adjacent in the sequence:

$$X = (x_1^v, x_1^\theta, x_2^v, x_2^\theta, \dots, x_{HW}^v, x_{HW}^\theta) \in \mathbb{R}^{2HW \times C'}. \quad (6.2)$$

The hidden state therefore passes from one modality to the other at every location. Fusion happens inside the state, not after it.

RELIABILITY-GATED SELECTIVE SCAN. The interleaved sequence is processed by a selective scan in the SS2D pattern [76], with one change. The step of each token is scaled by the reliability of its modality at its location:

$$\Delta_k = \text{softplus}(w_\Delta^\top x_k + b_\Delta) \cdot r_{p(k)}^{m(k)}, \quad (6.3)$$

where $m(k) \in \{v, \theta\}$ and $p(k)$ are the modality and location of token k . By the argument of Section 2.3.1, a token judged unreliable receives $\Delta_k \rightarrow 0$. It neither writes into the state nor erases it.

This differs in kind from the output gating of Fusion-Mamba and CFMW [29, 69]. Output gating suppresses a modality’s contribution after it has already updated the state. Gating the step prevents a corrupted token from contaminating the state in the first place. The difference is small on still images. It becomes decisive once the same state persists across frames.

MERGE. The scan output is de-interleaved into Y^v and Y^θ . A local branch, a depthwise 3×3 convolution on $[\phi_v(F^v), \phi_\theta(\tilde{F}^\theta)]$, runs alongside. This follows MambaVision’s finding that a non-SSM branch beside the scan helps [51]. The fused output is

$$Z = \hat{r}^v \odot F^v + \hat{r}^\theta \odot \tilde{F}^\theta + W_o[Y^v; Y^\theta; \text{local}], \quad \hat{r}^m = \frac{r^m}{r^v + r^\theta + \epsilon}, \quad (6.4)$$

which reduces to a reliability-weighted average when the scan contributes nothing. The block also outputs the combined reliability $\rho = \max(r^v, r^\theta)$, which the TSM uses.

COST. With d scan directions and state size N , the scan costs $\mathcal{O}(d \cdot 2HW \cdot C'N)$. At P_3 for a 640×512 input, $2HW = 10\,240$ tokens, which is well inside the regime where a linear scan is cheaper than attention. The number of directions ($d \in \{2, 4\}$) is an ablation.

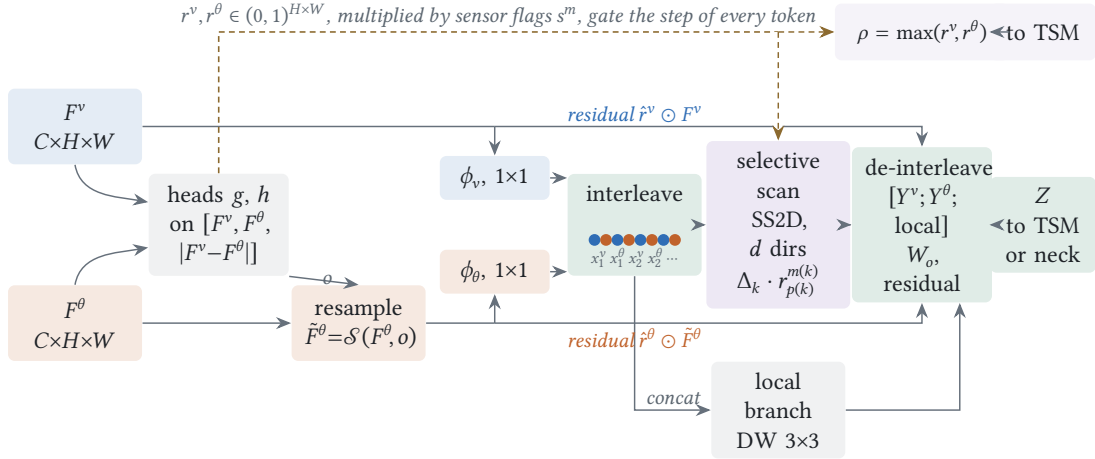


FIGURE 6.2. The Cross-Modal Fusion Mamba (CMFM) block at one pyramid level. Small heads estimate per-location reliability for each modality and a displacement field that resamples the thermal features onto the visible grid. The projected streams are interleaved so that the hidden state alternates between modalities at every location. The selective scan multiplies each token’s step Δ_k by the reliability of its modality and location (ochre path). An unreliable token therefore neither writes into the state nor erases it. A local convolutional branch and reliability-weighted residuals complete the block. The combined reliability ρ is passed to the Temporal State Memory.

6.5 Temporal State Memory

The TSM gives each location of each fused level a small state that persists across frames. For level l with resolution $H \times W$, the state is $S_t \in \mathbb{R}^{H \times W \times C_s \times N}$, with $C_s = C/2$ and $N = 8$ by default.

INPUT. The fused features are projected, $u_t = W_{\text{in}} Z_t \in \mathbb{R}^{H \times W \times C_s}$. The input-dependent projections $B_t = W_B u_t$ and $C_t = W_C u_t$ are formed as in Equation (2.4).

MOTION COMPENSATION. The previous state is warped into the current frame,

$$\hat{S}_{t-1} = \mathcal{W}(S_{t-1}; H_{t-1 \rightarrow t}), \quad (6.5)$$

by bilinear sampling at the level's resolution. Locations that enter the view have no history and receive a zero state.

GATED STEP. The temporal step combines the learned selection with the three control signals:

$$\Delta_t(p) = \text{softplus}(w^\top u_t(p) + b) \cdot \frac{\tau_t}{\tau_0} \cdot \rho_t(p). \quad (6.6)$$

Here τ_t is the measured interval since the previous processed frame and τ_0 the nominal interval seen in training. The factor τ_t/τ_0 keeps the discrete update faithful to the same continuous dynamics when frames are dropped or arrive irregularly. This is the property Zubić and colleagues exploited for event cameras [174]. The factor ρ_t holds the state where neither sensor can be trusted, so the memory of an object survives a gap in the feed.

UPDATE AND READOUT. For each location p ,

$$S_t(p) = \exp(\Delta_t(p) A) \odot \hat{S}_{t-1}(p) + \Delta_t(p) u_t(p) B_t(p)^\top, \quad (6.7)$$

$$y_t(p) = S_t(p) C_t(p) + D \odot u_t(p), \quad (6.8)$$

$$\tilde{Z}_t = Z_t + W_{\text{out}}(\text{DWConv}_{3 \times 3}(y_t)). \quad (6.9)$$

The depthwise convolution in Equation (6.9) spreads temporal evidence to neighbouring locations. That compensates for small errors in the warp. Because the TSM is residual and the state starts at zero, image mode is the special case of a single step from an empty state. CFFM-Net and CFFM-Net-S therefore share one set of weights.

RESET. The state is cleared in three cases:

- the stream monitor detects a scene cut;

- motion estimation fails its inlier test;
- the gap τ_t exceeds a maximum $\tau_{\max}$.

A soft reset, which sets Δ large, is an alternative evaluated in the ablation.

COST. Memory and computation per frame are $\mathcal{O}(HW C_s N)$ and do not depend on how many frames have been seen. At P_3 for a 640×512 input with $C_s = 64$ and $N = 8$, the state holds $80 \times 64 \times 64 \times 8 \approx 2.6$ million values, about 10 MB in 32-bit floats. All three levels together need about 14 MB. Figure 6.3 shows the memory unrolled over three frames, including a thermal dropout and a frame gap.

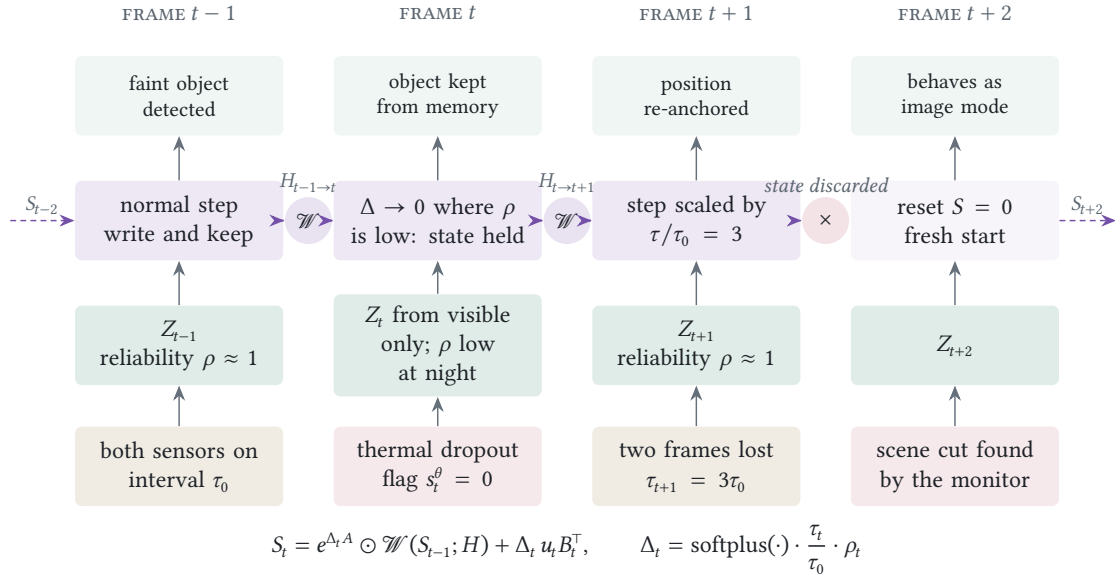


FIGURE 6.3. The Temporal State Memory unrolled over four frames that exercise its three controls. At $t - 1$ both sensors work and the state updates normally. At t the thermal sensor drops out at night. The reliability ρ falls where only thermal could see, the step Δ falls with it, and the state holds the object until the sensor returns. At $t + 1$ two frames have been lost, so the step is scaled by the true interval. At $t + 2$ the monitor detects a scene cut and the state is reset. Purple arrows carry the state through the motion warp $\mathcal{W}$.

6.6 Ego-motion compensation and scene changes

Drone and handheld footage moves. Without compensation, the state at a location would describe whatever was there before the camera moved.

- The default estimator is classical. ORB features are matched between consecutive frames at half resolution, and a homography is fitted with RANSAC. The thermal frame is used

at night and the visible frame by day, chosen by the stream monitor from the global reliability. The result is scaled to each pyramid level.

- For static cameras the estimator is bypassed and H is the identity.
- A learned alternative regresses the eight homography parameters from the P_5 features of frames $t - 1$ and t . It is cheaper on an embedded device and is compared in the ablation.

A homography models only a planar or distant scene. Parallax and independently moving objects are left to the learned update, the depthwise spreading in Equation (6.9) and the tracker.

6.7 Training

Training proceeds in two stages (Figure 6.4).

STAGE A, STILL IMAGES.

- Both backbones start from COCO-pretrained YOLO26 weights. The first convolution of the thermal backbone is initialised by averaging the pretrained filters over the colour channels.
- The CMFM blocks, neck and heads are trained on single frames from the visible-thermal datasets of Chapter 7.
- The result is the still-image CFFM-Net, which is also the Phase 1 deliverable.

STAGE B, CLIPS.

- Training continues on clips of L consecutive frames, with $L = 8$ and later $L = 16$.
- Each clip uses a random frame stride $s \in \{1, 2, 3\}$, so that the model sees varied intervals τ_t .
- The first two frames of each clip are a burn-in without loss. Gradients flow through the clip and are truncated at its boundary (truncated back-propagation through time [149]).
- With probability 0.5, the final state of one clip initialises the next clip of the same sequence. The model therefore experiences states that are much older than L frames.

AUGMENTATION AND CORRUPTION. Geometric augmentation is identical for both modalities and for every frame of a clip. Mosaic is disabled in Stage B because it breaks temporal continuity. Photometric jitter is applied to the visible stream only. Thermal frames receive gain and offset drift and sensor noise. A corruption simulator then applies five perturbations per clip.

- Bursty dropout of each modality is drawn from a two-state Markov chain, with probability 0.02 of failing and 0.2 of recovering per frame. Outages therefore last five frames on average.
- Rare total blackouts remove both sensors at once.
- Thermal misalignment is a random affine offset, up to 16 pixels of translation, $\pm 3^\circ$ of rotation and $\pm 5\%$ of scale. It drifts slowly within the clip.
- Desynchronisation delays the thermal stream by zero to two frames.
- Random frame drops make the intervals irregular.

Because the simulator knows what it corrupted, it also provides targets for the reliability heads.

OBJECTIVE. The loss averages the YOLO26 detection loss over the supervised frames of a clip, $\mathcal{T}$, and adds a weak reliability term:

$$\mathcal{L} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \mathcal{L}_{\text{det}}(t) + \lambda_{\text{rel}} \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \sum_{m \in \{v, \theta\}} \text{BCE}(\bar{r}_t^m, q_t^m), \quad \lambda_{\text{rel}} = 0.1. \quad (6.10)$$

Here $\bar{r}_t^m$ is the spatial mean of a modality’s reliability map and $q_t^m \in \{0, 1\}$ is the simulator’s target. Optimisation uses the YOLO26 defaults with mixed precision. The selective scan and the TSM state are kept in 32-bit precision, because T4 GPUs lack bfloat16 and the scan is sensitive to half-precision rounding. An exponential moving average of the weights is kept for evaluation.

6.8 Inference modes and complexity

The same weights run in three modes.

- *Image mode* uses an empty state for each frame. It is used in Phase 1 and as an internal baseline.
- *Clip mode* processes a stored video causally from its first frame. It is used in Phase 2. A bidirectional variant can be run offline as an upper bound.
- *Stream mode* keeps the state between calls and advances it by one step per arriving frame. It is used in Phases 3 and 4.

Table 6.1 compares the per-frame cost of the temporal options that the ablation will test.

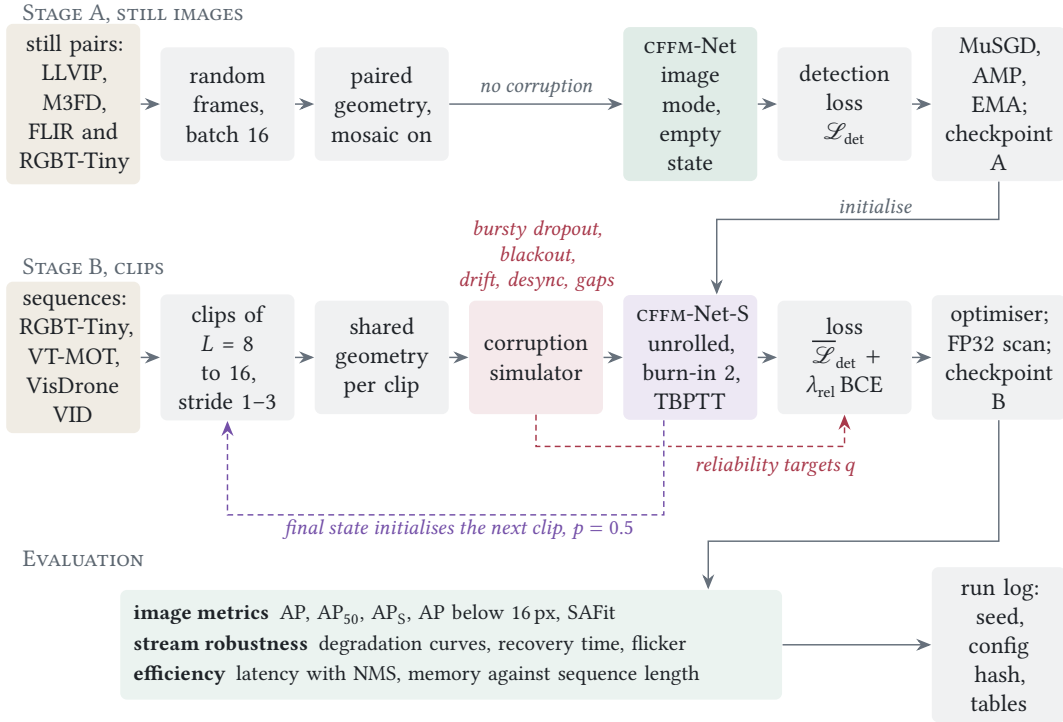


FIGURE 6.4. The two-stage training pipeline. Stage A trains the still-image CFFM-Net and yields the Phase 1 model. Stage B continues on clips with a corruption simulator, truncated back-propagation through time (TBPTT) and stochastic carry-over of the state between clips. Because the simulator knows which sensor it disabled or shifted, it provides targets for the reliability heads. Every evaluation is logged with its seed and configuration hash, and the tables of the final paper are generated from these logs.

TABLE 6.1. Per-frame cost of temporal designs as a function of the number of frames already seen, T , and the window length, k . M is the number of feature locations.

Design	Compute	Memory	Behaviour
per-frame detector	$\mathcal{O}(1)$	none	no temporal context
sliding window of k frames	$\mathcal{O}(k)$	$\mathcal{O}(kM)$	context limited to k frames
temporal attention over k	$\mathcal{O}(kM)$	$\mathcal{O}(kM)$	quadratic in k if all pairs attend
convolutional GRU	$\mathcal{O}(M)$	$\mathcal{O}(M)$	unbounded memory; gates not tied to time
Temporal State Memory	$\mathcal{O}(MN)$	$\mathcal{O}(MN)$	unbounded memory; step tied to reliability and true interval

6.9 Hypotheses and how they could fail

Each hypothesis is paired with the comparison that tests it and the outcome that would refute it (Table 6.2). The targets are deliberately modest. A refuted hypothesis is a publishable finding if the experiment is sound, so the protocol of Chapter 8 is designed to make every outcome interpretable.

TABLE 6.2. Hypotheses, the tests that address them and the outcomes that would refute them. “Seed spread” is the standard deviation over three training seeds.

	Hypothesis	Test	Refuted if
H1	Temporal memory improves tiny-object detection.	CFFM-Net-S against image-mode CFFM-Net on RGBT-Tiny, AP for objects under 16 px and SAFit.	The gain is below twice the seed spread.
H2	The selective state is the right memory.	TSM against convolutional GRU, a sliding window and windowed temporal attention at matched GFLOPs.	Any alternative matches TSM’s accuracy at lower per-frame latency.
H3	Gating the step by reliability gives graceful degradation.	Bursty thermal dropout and misalignment, against output gating, COMO-style fusion and concatenation.	Area under the degradation curve is not better, or recovery after an outage is not faster.
H4	Scaling the step by the true interval generalises across frame rates.	Train at 15 FPS without stride augmentation; test at 7.5 and 3.75 FPS and with random drops, with and without the factor τ_t/τ_0 .	Accuracy loss under rate shift is not reduced.
H5	The streaming model is deployable.	TensorRT FP16 export, desktop GPU and Jetson Orin, end-to-end latency including NMS and motion estimation.	Under 30 FPS on the desktop GPU or under 15 FPS on Orin at 640×512 , or more than 1 AP_{50} lost in export.

Datasets

The datasets decide what the paper can claim. This chapter sets out the criteria for choosing them and the recommended core stack. It describes the main benchmark in detail and lists the supporting datasets by role. It then covers licences, datasets that must not be used, and the data engineering that turns downloads into reproducible inputs.

7.1 Selection criteria

A dataset enters the plan only if it meets five criteria.

1. *Standing*. It is published at a recognised venue and used as a benchmark by the work the paper will be compared against.
2. *Relevance*. It contributes small objects, paired visible and thermal imagery, or temporal sequences. The best datasets contribute all three.
3. *Label quality*. Known label problems have published fixes, such as the corrected LLVIP labels and the sanitised KAIST labels.
4. *Licence*. It permits research use. All datasets used here permit it, and most forbid commercial use.
5. *Feasibility*. It can be stored, converted and trained on within the Kaggle and local budget of Section 8.7.

7.2 The core stack

Table 7.1 lists the eight datasets that every phase relies on. Together they cover still images and video, visible and thermal sensing, ground and aerial viewpoints, and the standard tracking metric.

TABLE 7.1. The core dataset stack, in order of first use. Licences are summarised; the official terms govern.

Dataset	Phase	Content	Role	Licence
COCO 2017 [70]	1	118 287 training images, 80 classes	pretrained weights; AP _s protocol	annotations CC BY 4.0
VisDrone2019-DET [171]	1	6 471 / 548 / 1 610 drone images, 10 classes	visible small objects	CC BY-NC-SA 3.0
LLVIP [60]	1	15 488 aligned pairs at 1280×1024 , person	first RGB-T pipeline test	non-commercial
M3FD [73] and FLIR-aligned [165]	1	4 200 pairs, 6 classes; 5 142 pairs, 3 classes	comparison with Fusion-Mamba, COMO	research use
DroneVehicle [131]	1	28 439 aerial pairs, 953 087 oriented boxes	aerial RGB-T, weak alignment	research use [†]
RGBT-Tiny [157]	1–3	115 paired sequences, 93 K frames, 1.2 M boxes	main benchmark	on request
VisDrone-VID and MOT17 [89, 171]	2	288 clips, 261 908 frames; 14 sequences	video detection, HOTA	non-commercial
MEVA [15]	3–4	328 h ground and UAV video with thermal, consented actors	surveillance-style stress test	CC BY 4.0

7.3 The main benchmark: RGBT-Tiny

RGBT-Tiny, published in IEEE TPAMI in 2025, is the single best match for this project [157].

- It contains 115 paired visible–thermal sequences: 93 000 frames and 1.2 million boxes in seven classes. The classes are ship, car, cyclist, pedestrian, bus, drone and plane.
- More than 81% of its objects are smaller than 16×16 pixels.
- The frames are 640×512 , aligned by homography, at 15 frames per second, across eight scene types.

- The official split is 85 training sequences (79 050 frames) and 30 test sequences (13 950 frames).
- Boxes carry track identities, and the companion VT-Tiny-MOT benchmark provides tracking labels on the same sequences [54]. Phases 1 to 3 can therefore all be evaluated on one data source.
- It proposes SAFit, a matching measure intended to be fairer than IoU for tiny boxes. SAFit is reported alongside standard AP.

Two cautions apply. Access is granted through a request form, so the request must be made at the start of Phase 0. Its alignment is by homography, so residual parallax remains at short range. That is a feature for testing the offset-guided alignment of the CMFM block. It is also a reason to audit the alignment before training (Section 7.7).

7.4 Supporting datasets

The supporting datasets fall into four groups by role. Table 7.2 covers small objects, Table 7.3 visible–thermal data, Table 7.4 video and tracking, and Table 7.5 general detection. Each table marks whether a dataset is in the plan (**core**), *optional*, or a *watch* item to reconsider before submission.

TABLE 7.2. Supporting datasets for small and tiny objects.

Dataset	Released	Content and notes	Status
AI-TOD v2 [154]	2022	28 036 images, 8 classes, mean size 12.8 px; built from xView and others; test labels withheld	optional
SODA-D, SODA-A [13]	2023	24 828 driving images; 2 513 aerial images with oriented boxes; CC BY-NC 4.0	optional
TinyPerson [162]	2020	1 610 images, 72 651 persons at sea and on land	optional
DOTA v1.5 [26, 150]	2019	2 806 images, 403 318 instances, very small objects labelled; academic use only	optional
xView [63]	2018	1 M objects, 60 classes, 0.3 m resolution; CC BY-NC-SA 4.0	source of AI-TOD
UAVDT [33]	2018	100 sequences, 80 K frames, vehicles, detection and tracking	optional
SeaDronesSee [139]	2022	maritime search and rescue; CC0, the only fully permissive set	optional
XS-VID [153]	2024	374 sequences, 223 K frames, extremely small objects in video	optional
TinySet-9M [133]	04/2026	multi-domain small-object set; details unconfirmed	watch

TABLE 7.3. Supporting datasets with paired visible and thermal (or near-infrared) imagery.

Dataset	Released	Content and notes	Status
KAIST [59, 66]	2015	95 K pairs of pedestrians; use the sanitised training and improved test labels; scored by miss rate	optional, for Mam-baST
VT-VOD50 [35]	2023	50 videos, 18 898 pairs, 7 classes, manually aligned	core for video
UVT-VOD2024 [93]	2024	about 60 K images, 11 classes, unaligned video	core for misalignment
VT-MOT [141]	2024	582 sequence pairs, 401 K frames, 3.99 M boxes	optional
VEDAI [106]	2016	1 210 aerial images, RGB and near-infrared, not thermal	optional
CVC-14 [44]	2016	far infrared with grayscale visible, poor alignment	cross-dataset only
M3OT [79]	2025	two drones, 10 790 RGB-IR pairs, multi-view vehicles	Phase 4
MMOT [90]	2025	eight-band multispectral tracking, not thermal	watch
DVT-VOD1000 [21]	07/2026	1 000 unaligned videos, 103 464 pairs; release announced	watch
DRGBT-1K [32]	07/2026	1 045 tracking sequences with modality switching	watch

TABLE 7.4. Supporting datasets for video detection and tracking.

Dataset	Released	Content and notes	Status
ImageNet VID [114]	2015	30 classes, 3 862 training snippets; standard video detection benchmark	optional
MOT20 [24]	2020	extreme crowds, up to 171 boxes per frame	optional
DanceTrack [130]	2022	100 videos, uniform appearance, nonlinear motion	optional
SportsMOT [16]	2023	240 clips of fast sports motion	optional
BDD100K MOT [161]	2020	2 000 driving videos	optional
TAO [20]	2020	2 907 videos, 833 classes, open vocabulary	optional
LaSOT [38]	2019	1 400 single-object tracking sequences	optional

TABLE 7.5. Supporting datasets for general detection and pretraining.

Dataset	Released	Content and notes	Status
COCO-ReM [125]	2024	corrected COCO masks and boxes for honest evaluation	optional
LVIS v1 [49]	2019	1 203 classes on COCO images, long tail	optional
Objects365 [121]	2019	1.74 M images; too large for this budget; use pretrained weights	weights only
Open Images V7 [62]	2022	1.74 M images with boxes; extract class subsets only	optional
PASCAL VOC [37]	2012	20 classes; fast debugging	debugging
RF100-VL [110]	2025	100 domains for generalisation checks	optional

7.5 Licences

Almost every dataset in the plan is licensed for non-commercial research only. That is sufficient for a thesis and a journal article. Four further rules follow from the terms.

- The data must not be redistributed. Kaggle datasets created from it must be private.
- The images of MSMT17 must not be shown in publications, at the authors' request.
- Any commercial follow-up would need datasets with permissive terms, such as SeaDronesSee (CC0) and MEVA (CC BY 4.0), or newly collected data.
- The Ultralytics library is licensed under AGPL-3.0. A public release of code built on it must use the same licence [160].

7.6 Datasets that must not be used

Several surveillance datasets were withdrawn after it emerged that people had been recorded without adequate consent (Table 7.6). Withdrawal did not stop their use. Peng, Mathur and Narayanan documented continued use of DukeMTMC and its derivatives in hundreds of papers after retraction [99]. A project inspired by a surveillance fiction must be scrupulous here.

TABLE 7.6. Withdrawn and restricted surveillance datasets.

Dataset	Status	Decision
DukeMTMC and derivatives	withdrawn in June 2019 after deviation from its ethics protocol; derivatives such as DukeMTMC-reID are equally affected	do not use
Brainwash	removed in June 2019 at the depositor’s request	do not use
MS-Celeb-1M	retracted in 2019	do not use
Oxford Town Centre	flagged as recorded without consent	avoid
MSMT17	available by agreement; images must not appear in publications	only if essential
Market-1501, CUHK03	collected without documented consent; face-blurred versions exist	legacy comparison only
MEVA	recorded with consenting actors, CC BY 4.0	preferred alternative

7.7 Data engineering

Figure 7.1 shows the path from downloads to training inputs. Five practices make the results reproducible.

IMMUTABLE RAW DATA. Archives are stored as downloaded, with checksums recorded in a dataset card. Every later step is a script that reads raw data and writes derived data. Derived data can always be regenerated.

ONE PAIRED FORMAT. Visible and thermal frames are stored in two folders with identical file names. A single label file per frame is expressed in the visible frame’s coordinates. Ultralytics’ multi-channel input stacks the channels into one TIFF and is convenient for early fusion [160]. The two-folder layout is used for middle fusion because the backbones are separate.

ALIGNMENT AUDIT. For each visible–thermal dataset, the residual offset between modalities is measured on a sample of frames. The measurement uses phase correlation of edge maps, and the distribution is reported. The audit tells the paper how aligned each benchmark really is. That calibrates the synthetic misalignment used in training and evaluation.

LEAKAGE-FREE SPLITS. Frames from one video are strongly correlated. Splits are therefore made at the sequence level, and official splits are used wherever they exist. No sequence may contribute frames to both training and test sets.

PACKAGING FOR KAGGLE. Converted datasets are uploaded as private Kaggle datasets, versioned and shared across notebooks, so that each training session starts without re-downloading. Ignore regions, such as VisDrone’s, are carried into evaluation and not silently dropped.

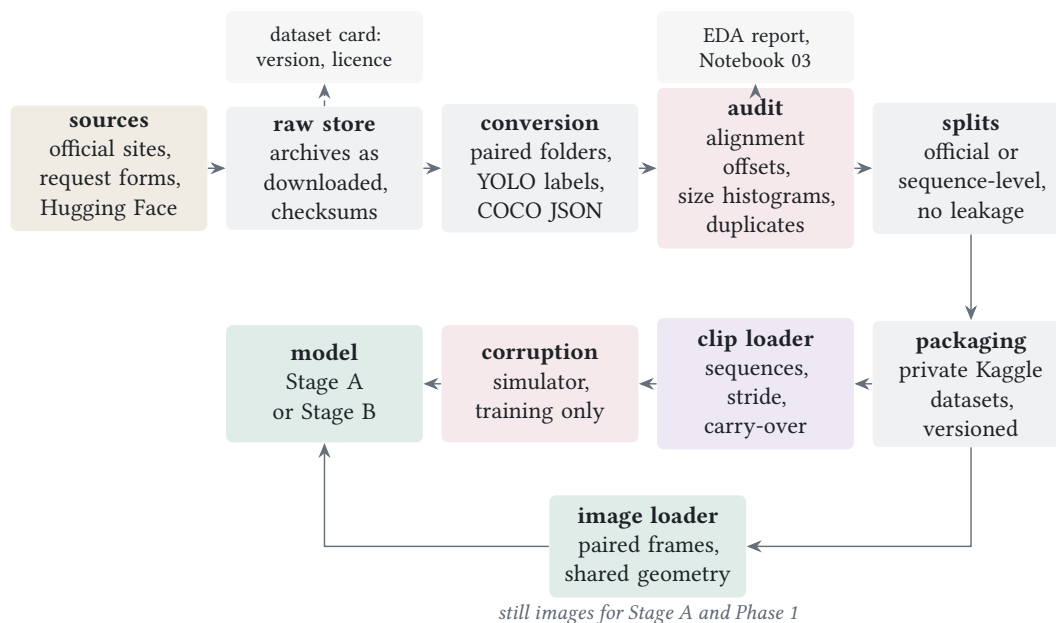


FIGURE 7.1. The data engineering pipeline. Raw archives are never modified. Every derived artefact is produced by a script and recorded in a dataset card. The audit measures the real alignment of each visible–thermal benchmark before any training. Converted data is packaged once as private Kaggle datasets and consumed by the image and clip loaders.

Experimental Protocol

The protocol is written before any experiment runs, so that the results cannot shape the questions. It specifies the metrics, the baselines, the ablations, the robustness tests, the statistical treatment, the latency measurement and the compute budget. Each choice answers a demand that reviewers at the target journals make routinely (Section 12.4).

8.1 Metrics

STILL IMAGES. COCO AP, AP₅₀ and AP₇₅ are computed with `pycocotools` or the faster compatible `faster-coco-eval` [39, 70]. Ultralytics reports only AP₅₀ and AP, so predictions are exported as COCO JSON and scored externally. Small-object performance is reported in three ways:

- the COCO size bands;
- the AI-TOD bands of 2–8, 8–16 and 16–32 pixels;
- SAFit-based AP on RGBT-Tiny [157].

If KAIST is used, the log-average miss rate is also reported.

VIDEO AND TRACKING. Video detection is scored per frame over whole sequences. Tracking uses TrackEval [77, 136], which reports HOTA with its detection and association components, IDF1 and MOTA. Temporal stability is measured as *flicker*. Flicker is the fraction of ground-truth track frames in which a detected object is lost and found again within three frames.

STREAMING ROBUSTNESS. Each perturbation of Section 8.4 is applied at several severities. The result is a degradation curve of AP against severity. Its normalised area (AUDC) summarises robustness in one number. Two event metrics describe behaviour around outages.

- *Hold recall* is the recall, during an outage, of objects that were visible before it began.
- *Recovery time* R_{95} is the number of frames after a sensor returns until per-frame recall reaches 95% of the recall on the same frames without the outage.

EFFICIENCY. The efficiency metrics are parameters and GFLOPs at 640×512 , and end-to-end latency at batch one (Section 8.6). They also include throughput, peak GPU memory, and memory as a function of sequence length. For CFFM-Net-S, the memory curve should be flat.

8.2 Baselines

Every baseline is retrained under the same schedule, input size and evaluation code. None are copied from papers. Where official code exists it is used; otherwise the method is reimplemented and the differences are stated. Table 8.1 lists the four groups.

TABLE 8.1. Baselines by group. The code column records whether an official implementation is expected to be available; reimplementations are labelled as such in the paper.

Group	Methods	Code
single modality, per frame	YOLO26-s on visible; YOLO26-s on thermal; RF-DETR-S on visible as a DETR reference [111]	official
static fusion	two-stream YOLO26 with concatenation and addition; CFT-style attention fusion [105]; Fusion-Mamba [29]; COMO [14]; WaveMamba [148]; image-mode CFFM-Net	mixed; reimplement where missing
temporal	sliding-window feature aggregation with $k \in \{3, 5, 9\}$; convolutional GRU [5]; windowed temporal attention; three-frame MambaST-style fuser [86]; bidirectional temporal SSM as an offline upper bound; MSENNet or MSGNet if released [93, 94]	mostly reimplemented
tracking	ByteTrack, BoT-SORT and TrackTrack on top of each detector [2, 123, 166]	official, in Ultralytics

The temporal baselines are the critical group. Each receives the same per-frame fused features Z_t as the TSM and replaces only the temporal module. The comparison therefore isolates the memory mechanism. This is the control that MambaOut-style critiques demand [163]. It is complemented by a *gated-convolution control*: the TSM with its state-space recurrence replaced by a gated convolution of matched cost.

8.3 Ablations

Table 8.2 lists the ablations, each tied to a design decision of Chapter 6. To fit the compute budget, every ablation first runs with one seed. Those that change the conclusion of a hypothesis are repeated with three seeds.

TABLE 8.2. The ablation plan. The default setting is listed first in each row.

	Component	Variants	Tests
A1	stride-4 head	on; off	P1
A2	early capacity	UAVDet-style; YOLO26 default; intermediate	P1
A3	fusion operator	CMFM with gated step; CMFM with output gating; attention; concatenation; addition	H3
A4	token order	interleaved; separate scans then merge	H3
A5	scan directions	four; two	cost
A6	offset alignment	on; off	H3
A7	TSM placement	P_3-P_5 ; P_3 only; P_3-P_4 ; including P_2	H1
A8	state size N	8; 4; 16	H1
A9	motion compensation	classical homography; learned; none	H1
A10	interval scaling τ_t/τ_0	on; off	H4
A11	reliability supervision	on; off	H3
A12	clip training	$L = 16$ with carry 0.5; $L = 8$; no carry	H1
A13	reset	hard; soft; none	H1
A14	memory mechanism	TSM; gated convolution; ConvGRU; temporal attention	H2
A15	backbones	separate; shared from stage 3	cost

8.4 Robustness protocol

No benchmark protocol exists for temporal sensor failure in visible–thermal detection (Section 5.2). This section defines one, and it is itself a contribution of the paper (Table 8.3). All perturbations are applied at test time to the RGBT-Tiny test sequences with fixed random seeds, so that every method sees identical corrupted streams. Real misalignment is tested on UVT-VOD2024 and on the shifted subset of DroneVehicle [7, 93].

TABLE 8.3. The streaming robustness protocol. Severities are applied separately unless combined in the last row.

Perturbation	Parameters and severities	Metrics
independent dropout	each modality dropped per frame with probability 0.1, 0.3, 0.5	AP, AUDC
bursty dropout	Markov outages with mean length 5, 15, 45 frames	hold recall, R_{95}
blackout	both sensors absent for 5 or 15 frames	hold recall, R_{95}
misalignment	thermal translation 0, 4, 8, 16 px; rotation 0, 1, 3 degrees; scale 0, 2, 5%	AP, AUDC
drift	offset growing linearly to the maximum over 300 frames	AP over time
desynchronisation	thermal delayed by 0, 1, 2 frames	AP
frame rate	strides 2 and 4 (7.5 and 3.75 FPS) and random drops of 10% and 30%	AP, flicker
sensor degradation	thermal gain drift and noise; visible low light and synthetic fog	AP
combined	bursty thermal dropout with 8 px misalignment and 10% frame drops	all

8.5 Statistical practice

- Every main configuration is trained with three seeds and reported as mean and standard deviation.
- Frames within a sequence are not independent, so confidence intervals are computed by bootstrap over *sequences*, with 10 000 resamples.
- Comparisons between two methods use the paired bootstrap of their difference on the same resampled sequences. A difference is reported as significant when its 95% interval excludes zero.
- The five hypotheses form a family. Holm’s correction is applied to their tests.
- Effect sizes are reported in AP points, not only as significance.

8.6 Latency and efficiency measurement

Latency is measured by one script for every model, on the same machine, with the software versions recorded. The protocol follows the cautions of RF-DETR [111].

- The input is batch one at 640×512 , with FP16 TensorRT engines.

- Warm-up runs 50 iterations. Measurement runs 500 iterations, with a 200 ms pause between them to avoid thermal throttling. The median and the 95th percentile are reported.
- The timed region includes pre-processing, the network, NMS where the head needs it, motion estimation and the TSM update.
- Accuracy is measured on the exported engine, not on the PyTorch model, so that any loss in export is visible.
- On Jetson Orin, the power mode and clock settings are stated.

8.7 Compute budget

The work runs on two resources (Table 8.4). A local laptop GPU handles development, testing and live demonstrations. Kaggle notebooks with T4 GPUs handle training. Kaggle runs Linux, so the mamba_ssm kernels can be compiled there. The compiled wheel is stored as a private Kaggle dataset so that later sessions install it in seconds. The T4 has compute capability 7.5 and no bfloat16. Kernel support on it must be verified in Notebook 00, with the pure-PyTorch scan as the fallback [83].

TABLE 8.4. Compute resources and their roles. Kaggle limits change over time and must be checked when the work starts.

Resource	Specification	Role
local laptop	RTX 4050 Laptop GPU, 6 GB, Windows 11; WSL2 Ubuntu for Linux tools	unit tests, debugging at small scale, visualisation, live demonstration, laptop latency
Kaggle notebooks	two T4 GPUs with 16 GB each, or one P100; sessions of up to 12 h; weekly GPU quota of about 30 h [†]	all training and evaluation runs; distributed data parallel over two T4s
optional	university GPU cluster, Colab, a borrowed Jetson Orin	contingency for quota; edge latency for H5

Table 8.5 estimates the training budget in T4 GPU-hours. The estimates rest on three assumptions, which Notebook 04 replaces with measurements:

- the small (s) model scale;
- native 640×512 input with COCO-pretrained backbones;
- RGBT-Tiny training frames subsampled by three, which loses little because neighbouring frames at 15 FPS are nearly identical.

About 440 T4-hours fit into roughly fifteen weeks of Kaggle quota, which is compatible with the timeline in Chapter 12. Running sessions continuously and using both T4s per session reduces wall-clock time.

TABLE 8.5. Estimated training budget in T4 GPU-hours. All figures are planning estimates to be replaced by measurements.

Experiment group	Configurations	Seeds	Hours each	Total
Phase 1 baselines and CFFM-Net on RGBT-Tiny	7	3	4	84
image ablations A1–A6, A15	10	1	4	40
confirmation of decisive image ablations	3	2	4	24
Stage B: CFFM-Net-S and temporal baselines	6	3	6	108
temporal ablations A7–A14	10	1	6	60
other datasets: VisDrone, DroneVehicle, VT-VOD50, UVT-VOD2024, KAIST	15	1–3	4	90
robustness evaluation, tracking, export				30
<i>total</i>				<i>436</i>

Kaggle sessions end after at most twelve hours. Every training script therefore writes a checkpoint at the end of each epoch and resumes automatically from the latest one. Runs are sized to finish within a session wherever possible.

Phased System Roadmap

The system is built in five phases. Phase 0 prepares the environment and data. Phases 1 to 4 each deliver something usable on its own: a still-image detector, a video analyser, a live camera system and anonymous multi-camera analytics. Each phase ends with an exit test. The next phase starts only when the test passes. Figure 9.1 summarises the sequence.

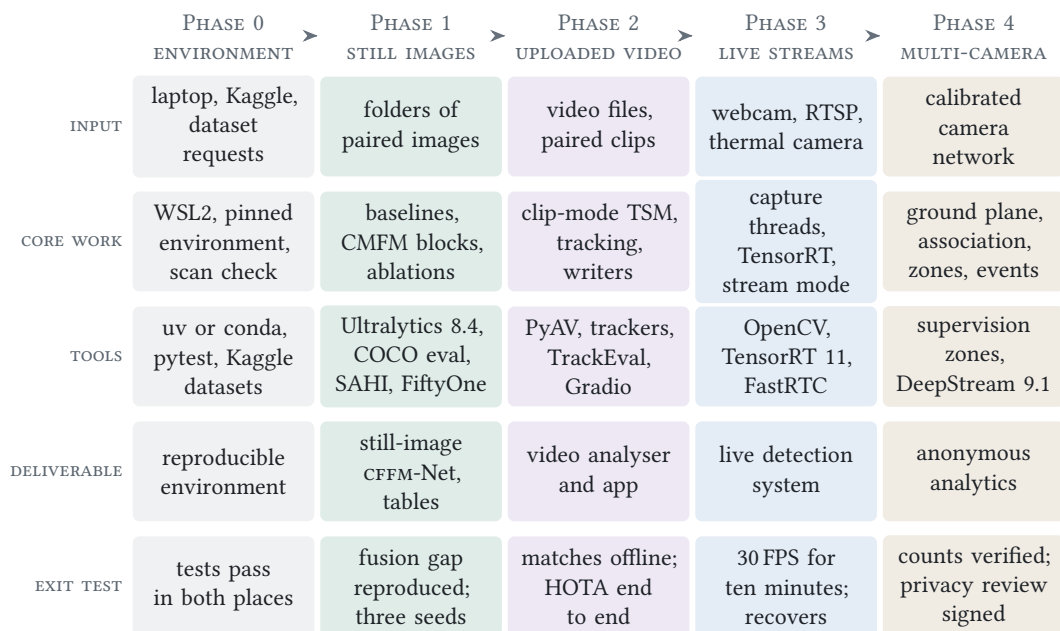


FIGURE 9.1. The five phases of the build. Each column is a phase and each row one aspect of it. A phase begins only when the exit test of the previous phase has passed. Phases 1 to 3 share one set of weights. Phase 4 is optional and bounded by the privacy rules of Chapter 11.

9.1 Phase 0: environment and data access

Phase 0 removes the obstacles that stall projects later.

LOCAL MACHINE. On the Windows laptop, WSL2 with Ubuntu 24.04 provides a Linux environment. GPU access comes from the Windows NVIDIA driver; no Linux driver is installed inside WSL. Datasets are kept on the Linux file system, not under `/mnt/c`, because cross-system file access is slow. A single Python environment pins PyTorch, Ultralytics 8.4, the evaluation tools and the video libraries. Jupyter runs inside WSL and is opened from the Windows browser or from VS Code.

KAGGLE. A setup notebook installs the same pinned environment and compiles `mamba_ssm` once. It saves the wheel as a private dataset and verifies the kernels on the T4. It also checks that the pure-PyTorch scan agrees with the CUDA scan.

DATA REQUESTS. Requests for RGBT-Tiny, LLVIP, FLIR ADAS and KAIST are sent on the first day, because approval can take weeks.

EXIT TEST. Phase 0 is complete when four checks pass.

- The unit tests pass both locally and on Kaggle.
- The two scan implementations agree to within 10^{-4} .
- A small YOLO26 trains for one epoch on a VisDrone subset in both places.
- The first datasets are packaged.

9.2 Phase 1: still images

Phase 1 delivers the still-image CFM-Net, the baselines it is compared with, and a tool that runs any of them on a folder of images. Figure 9.2 shows the pipeline. Images are read in pairs and letterboxed to the model’s input. For large aerial images, they are optionally sliced with SAHI [4]. Predictions are written as COCO JSON and scored externally, and the results are inspected visually in FiftyOne.

The order of work is set by risk.

1. Single-modality YOLO26 baselines are trained first. They test the whole pipeline on a known model.
2. Simple two-stream fusion follows.

3. The CMFM block is added last, once the pipeline around it is trusted.

EXIT TEST. The two-stream baseline must reproduce the published gap between single-modality and fused detection on LLVIP or M3FD within two points. The still-image CFFM-Net must then complete three seeds on RGBT-Tiny, with results generated by Notebook 15.

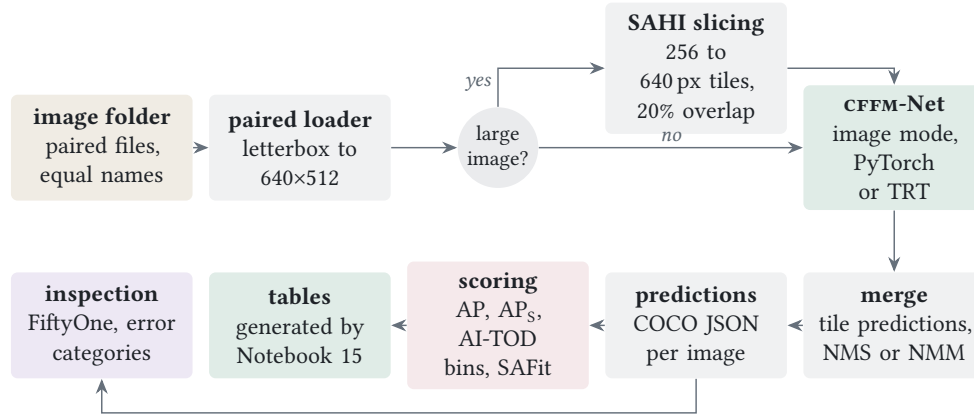


FIGURE 9.2. Phase 1 pipeline for still images. Large images are optionally sliced, and tile predictions are merged before scoring. Predictions are exported as COCO JSON and scored with external tools, so that size-specific and SAFit metrics are available. Errors are inspected and categorised in FiftyOne before any change to the model.

9.3 Phase 2: uploaded video

Phase 2 turns the detector into a video analyser. A user uploads a video, or a pair of visible and thermal videos. The system returns an annotated video, a track file and summary statistics (Figure 9.3).

DECODING. Videos are decoded with PyAV, which ships Windows wheels with FFmpeg included, or with OpenCV [104]. The decord library is avoided, because its last release dates from 2021. Paired videos are matched frame by frame using their timestamps. Each thermal frame is assigned to the nearest visible frame within half a frame interval. Unmatched frames are recorded as dropouts for the stream monitor, not silently discarded.

CLIP-MODE INFERENCE. CFFM-Net-S runs causally from the first frame and carries its state through the video. The same video can be re-run in image mode. This gives a direct, per-video comparison of what the memory contributes.

TRACKING. Detections are linked into tracks with the trackers built into Ultralytics (ByteTrack, BoT-SORT and TrackTrack) or with the Apache-licensed Roboflow trackers package [112]. Camera-motion compensation in BoT-SORT matters for drone footage. Tracks are written in MOTChallenge format and scored with TrackEval.

INTERFACE. A Gradio application accepts uploads and shows the annotated result beside the per-frame latency [45].

EXIT TEST. On the RGBT-Tiny and VT-Tiny-MOT test sequences, the pipeline must produce the same detections as the offline evaluation code. Its tracking metrics must be computed end to end by TrackEval.

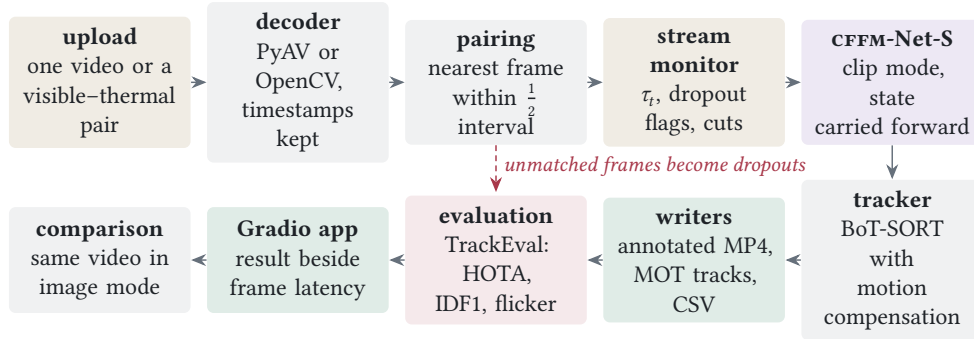


FIGURE 9.3. Phase 2 pipeline for uploaded video. Decoding keeps timestamps so that visible and thermal frames can be paired and the true interval measured. Unmatched frames are recorded as dropouts, not discarded. The streaming model runs causally from the first frame. The same video is re-run in image mode to show what the memory adds.

9.4 Phase 3: live camera streams

Phase 3 connects the streaming model to cameras: a webcam, an IP camera over RTSP, and a USB or network thermal camera. The difficulty here is not accuracy but time. Figure 9.4 shows the threaded design.

CAPTURE. Each camera has its own capture thread. On Windows, webcams are opened through the DirectShow or Media Foundation back ends. RTSP streams are read over TCP to avoid packet loss. Each thread writes into a queue that holds a single frame: the newest frame replaces the old one, so the model always sees the present and never a backlog. Every frame carries its capture timestamp. The stream monitor pairs visible and thermal frames by timestamp and computes the true interval τ_t . It raises the sensor flags when a camera stalls.

INFERENCE. The inference worker holds CFFM-Net-S as a TensorRT FP16 engine in stream mode. Its state persists between calls. Motion estimation runs in a separate CPU thread on downscaled frames, in parallel with the network. The tracker and renderer run in a third thread. The annotated stream reaches the browser by WebRTC through FastRTC, or by MJPEG on a local network [40].

BUDGET. At 30 frames per second each frame has about 33 ms (Table 9.1). Because the stages run in parallel, throughput is limited by the slowest stage, not by the sum. End-to-end delay, from light entering the camera to pixels on the screen, is the sum. The target is under 150 ms. When the model cannot keep up, three strategies apply, in this order.

1. Frames are dropped at capture, never in the middle of the pipeline.
2. The interval factor τ_t/τ_0 tells the TSM how much time has passed.
3. The detector can run every second frame, with the tracker predicting in between.

TABLE 9.1. Latency budget for live operation at 30 frames per second. Values other than the YOLO26 reference are planning estimates to be measured in Notebook 13.

Stage	Estimate (ms)	Notes
capture and decode	5–10	per camera thread, in parallel
pre-processing	1–3	letterbox and normalise on the GPU
network, two backbones, CMFM and TSM	8–15	YOLO26-s alone is 2.5 ms on a T4 [160]
NMS	0–1	none with the one-to-one head
motion estimation	2–4	CPU thread, overlaps the network
tracking	1–3	BoT-SORT or ByteTrack
rendering and encoding	5–10	overlay and WebRTC encoding
network to browser	20–60	local network or internet
<i>end to end</i>	<i>under 150</i>	<i>target</i>

EXIT TEST. On the desktop GPU, the live system must sustain 30 frames per second at 640×512 with paired cameras for ten minutes, without growth in delay or memory. It must also recover correctly when a camera is unplugged and reconnected. The same test on Jetson Orin at 15 frames per second answers H5.

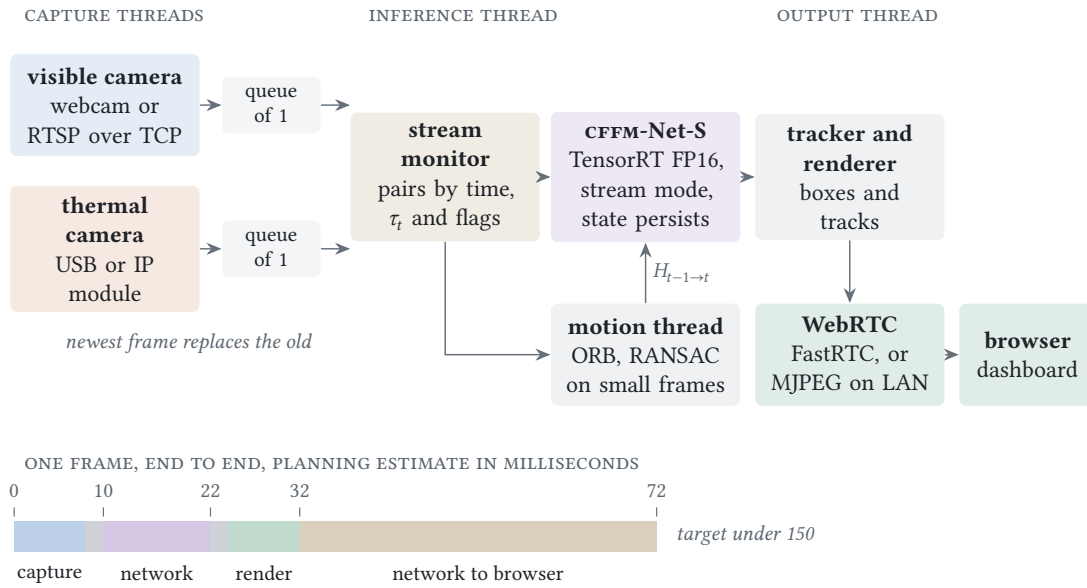


FIGURE 9.4. Phase 3 threaded design for live streams. Each camera has a capture thread writing into a queue of one frame, so the model always processes the present. The stream monitor pairs frames by timestamp and supplies τ_t and the sensor flags. Motion estimation runs in its own thread, in parallel with the network. The strip below shows a planning estimate of where the time goes for one frame. Stages run concurrently, so throughput is set by the slowest stage and delay by their sum.

9.5 Phase 4: anonymous multi-camera analytics

Phase 4 is optional. It is the closest the project comes to the fictional Machine, and it is deliberately limited (Chapter 11). Several calibrated cameras each run Phase 3. Their tracks are projected onto a common ground plane through camera-to-ground homographies. Tracks are associated across cameras by position, time and motion. For vehicles, non-biometric appearance such as colour and shape can be added. Person tracks are never associated through faces or other biometric features. The outputs are aggregate and anonymous:

- counts per zone;
- line crossings;
- dwell times;
- heat maps;
- rule-based events such as intrusion into a restricted area or a vehicle travelling the wrong way.

On NVIDIA hardware, DeepStream 9.1 offers multi-camera pipelines and three-dimensional tracking that can replace the custom orchestration [22]. Figure 9.5 shows the design.

EXIT TEST. Faces are blurred at ingest. No identity or face embedding is stored, and recordings are deleted on schedule. The counts must agree with manual counts on a held-out recording. The privacy review of Section 11.3 must be signed off.

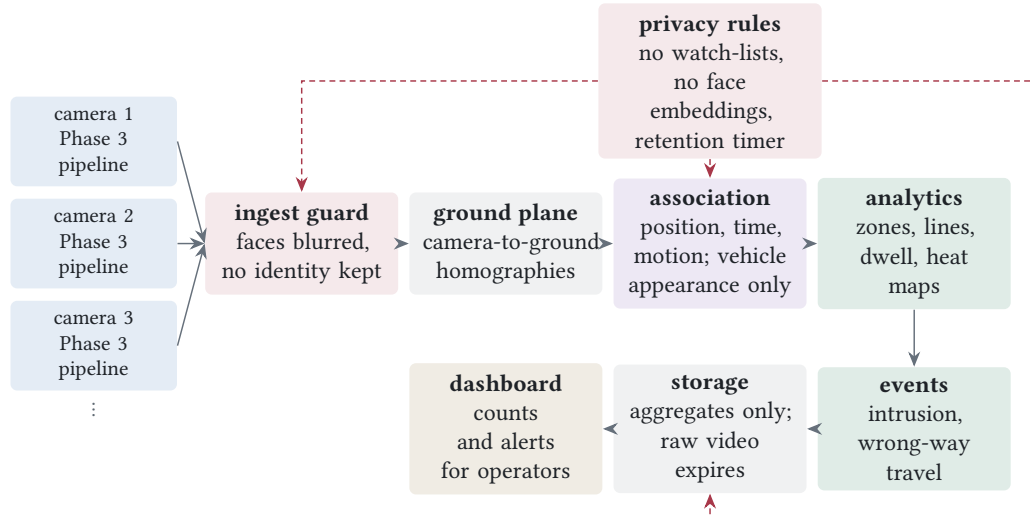


FIGURE 9.5. Phase 4 anonymous multi-camera analytics. Every camera runs the Phase 3 pipeline. An ingest guard blurs faces and discards identity before anything is stored. Tracks are associated across cameras on a shared ground plane by position, time and motion. Appearance cues are used only for vehicles. The outputs are aggregates and rule-based events. The privacy rules (red, dashed) constrain ingest, association and storage.

9.6 Exit criteria at a glance

Table 9.2 collects the exit test and the deliverable of every phase in one place.

TABLE 9.2. Exit criteria for each phase.

Phase	Exit test	Deliverable
0, environment	tests pass locally and on Kaggle; scans agree to 10^{-4} ; data packaged	reproducible environment
1, still images	published fusion gap reproduced within 2 points; CFM-Net with three seeds on RGBT-Tiny	image detector and tables
2, uploaded video	identical detections to offline evaluation; TrackEval end to end	video analyser and app
3, live streams	30 FPS for ten minutes without drift; recovery from unplugging; Orin at 15 FPS	live system
4, multi-camera	counts match manual counts; privacy review signed	anonymous analytics

Notebook Workflow and Repository Design

The work is carried out in Jupyter notebooks. Notebooks are good for exploration and poor at reproducibility. The design below keeps their convenience and removes their usual failure modes.

10.1 Principles

1. *Notebooks drive; modules hold the code.* Every model, loss, dataset and metric lives in a tested Python package. Notebooks import it, set parameters, run and display. Nothing a result depends on is defined only inside a notebook cell.
2. *One notebook, two machines.* Each notebook detects whether it runs locally or on Kaggle and sets paths accordingly. The same file runs in both places without edits.
3. *Configuration as files.* Data, model, training and evaluation settings are YAML files under version control. A run is identified by the hash of its resolved configuration.
4. *Every run is logged.* Each run records five things:
 - its seed;
 - the git commit;
 - the configuration hash;
 - the library versions;
 - its metrics, written as machine-readable JSON.

5. *Results are generated, never typed.* The tables and figures of the paper are produced from the run logs by Notebook 15.

10.2 Repository layout

cffm-net/	
pyproject.toml	pinned dependencies
configs/	
data/	rgbt_tiny.yaml, llvip.yaml, m3fd.yaml, ...
model/	yolo26s_dual.yaml, cffm_net_s.yaml, ablations/
train/	stage_a.yaml, stage_b.yaml
eval/	robustness.yaml, latency.yaml
src/cffm/	
data/	paired and clip datasets, corruption simulator, converters, alignment audit
models/	backbones, cmfm.py, scan.py (CUDA and PyTorch), tsm.py, egomotion.py, monitor.py, registry.py
train/	stage_a.py, stage_b.py, losses.py, tbptt.py
eval/	coco_eval.py, size_bins.py, safit.py, stream_metrics.py, track_eval.py, latency.py, bootstrap.py
deploy/	export_trt.py, stream_runtime.py, capture.py, webrtc_app.py
utils/	logging, seeding, checkpoint and resume
notebooks/	00 to 15, described in Table 10.1
tests/	unit, property and regression tests
scripts/	kaggle_setup.sh, build_mamba_wheel.sh, package_dataset.py
runs/	run logs (synchronised, not committed)
paper/	manuscript with generated tables

The model modules are registered with Ultralytics’ model parser, so a model is defined by a YAML file in the Ultralytics format and trained with its trainer. A custom trainer subclass adds paired inputs, clip training and the reliability loss.

10.3 The notebook sequence

Table 10.1 lists the sixteen notebooks, and Figure 10.1 shows how they depend on one another.

TABLE 10.1. The sixteen notebooks. The location column says where each normally runs: L is the local laptop, K is Kaggle.

No.	Notebook	Purpose and main output	Runs
00	environment setup	install, GPU check, compile and verify mamba_ssm, scan equivalence, smoke training	L, K
01	data acquisition	downloads, request tracking, checksums, dataset cards	L
02	data conversion	paired folders, YOLO labels, COCO JSON, ignore regions	L, K
03	data audit	size histograms, density, class balance, alignment offsets	K
04	single-modality base-lines	YOLO26 on visible and thermal; measured throughput replaces estimates	K
05	fusion baselines	two-stream fusion, CFT-style, Fusion-Mamba, COMO	K
06	CMFM tests	gating, interleaving and reliability maps, visualised	L
07	Stage A training	still-image CFM-Net with three seeds	K
08	image ablations	A1 to A6 and A15	K
09	clips and TSM	clip loader, corruption simulator checks, constant-memory test	L, K
10	Stage B training	CFM-Net-S and temporal baselines	K
11	robustness evaluation	the protocol of Table 8.3	K
12	tracking evaluation	ByteTrack, BoT-SORT, TrackTrack with TrackEval	K
13	export and latency	TensorRT export, latency protocol, accuracy of the engine	L
14	live demonstration	capture threads, stream runtime, FastRTC application	L
15	results and tables	aggregation, bootstrap intervals, LaTeX tables and figures	L

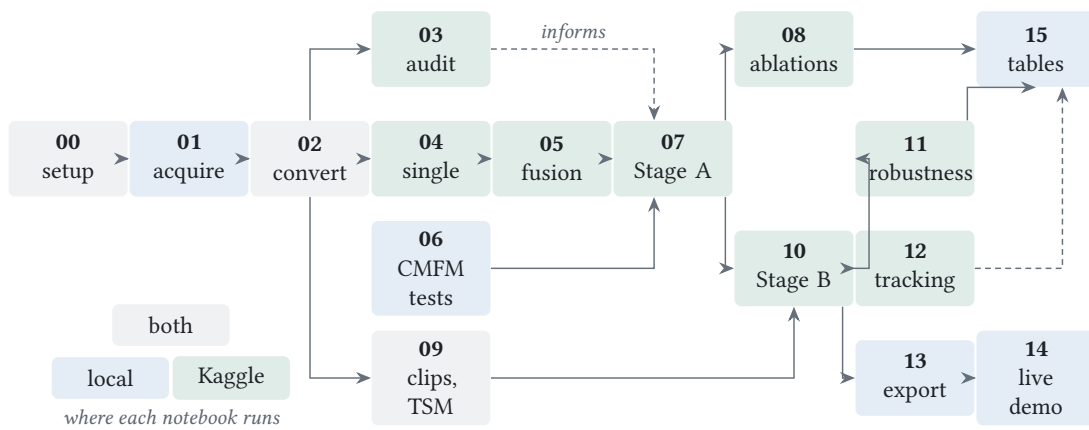


FIGURE 10.1. Dependencies between the notebooks. Colour shows where each notebook normally runs. Every training and evaluation notebook writes run logs, and Notebook 15 reads all of them. Only the main dependencies into it are drawn.

10.4 Testing

Three kinds of test protect the results.

- *Unit tests* check shapes and numerical agreement. The most important check is that the CUDA and pure-PyTorch scans agree.
- *Property tests* check the behaviour the method relies on.
 - With $\Delta = 0$ the state does not change.
 - With zero reliability for a modality, its tokens do not alter the state.
 - Image mode equals a single step from an empty state.
 - Memory use of the TSM does not grow with sequence length.
- *Regression tests* run a tiny model on a fixed fixture of frames. They fail if its outputs change unexpectedly.

The tests run in Notebook 00 on both machines and before every training run.

10.5 Generating results

Notebook 15 is the only route from experiments to the manuscript.

1. It reads every run log and groups runs by configuration.
2. It computes means, standard deviations and sequence-level bootstrap intervals.
3. It applies the multiple-comparison correction.
4. It writes LaTeX tables and plot data into the paper's directory.

A number in the paper can therefore always be traced to the runs that produced it. A changed run changes the paper on the next build, and no table is edited by hand.

Ethics, Law and the Person of Interest

Question

The project was inspired by a fictional system of mass surveillance. That inspiration is a reason for more care, not less. This chapter states what the system will not do and why. It summarises the legal framework that applies in India and the European Union, sets out the design rules that follow, and drafts the ethics statement the paper will carry.

11.1 What the Machine does that this system will not

The Machine of *Person of Interest* combines three capabilities beyond detection [101].

- It *identifies* individuals and follows them across cameras.
- It *profiles* them from their movements and associations.
- It *predicts* who will commit or suffer a crime.

Each of these is, in law, a distinct and heavily regulated activity: remote biometric identification, profiling and predictive risk assessment.

The system in this dossier is a perception layer. It detects and tracks objects, and it tracks people only as anonymous moving boxes. It does not recognise faces, it builds no identities and it makes no predictions about individuals. The gap between the two is not a technical limitation waiting to be closed. It is a boundary the project keeps on purpose.

11.2 The legal framework

EUROPEAN UNION. The AI Act, Regulation (EU) 2024/1689, has prohibited a set of practices since 2 February 2025 [36]. Article 5 bans four that matter here.

- Assessing a person’s risk of committing a criminal offence based solely on profiling or personality traits.
- Building facial-recognition databases by untargeted scraping of images from the internet or CCTV.
- Inferring sensitive characteristics, such as political opinion or religion, from biometric data.
- Real-time remote biometric identification in publicly accessible spaces for law enforcement, apart from three narrow exceptions that each require prior authorisation.

Other remote biometric identification is classed as high-risk. The application date of those obligations was postponed to December 2027 by later amendment[†]. Systems developed solely for scientific research are outside the Act’s scope, but their deployment is not.

The General Data Protection Regulation adds two further constraints. It treats biometric data used for unique identification as a special category. It requires an impact assessment before systematic monitoring of publicly accessible areas on a large scale [42].

INDIA. The Digital Personal Data Protection Act, 2023, and its Rules, notified in November 2025, come into force in phases. The core obligations apply from May 2027[†] [30, 31]. Three provisions matter here.

- Processing personal data, including images of identifiable people, requires a lawful basis, normally consent.
- The exemption for research and statistical purposes applies only when no decision is taken about a specific individual and prescribed safeguards are met.
- Tracking or behavioural monitoring of children is prohibited.

Beneath the Act sits the Supreme Court’s recognition of privacy as a fundamental right in *Puttaswamy*. Under that judgment, any surveillance must be lawful, necessary and proportionate [103].

11.3 Design rules

The rules below apply in every phase. They are written so that a reviewer, an ethics committee or a future user can check them.

1. No face recognition, re-identification of people, watch-lists or identity databases are built or used.
2. Faces in any footage the project records are blurred at ingest, before storage.
3. Processing happens on the device where possible. Only detections and aggregates leave it.
4. Phase 4 outputs are aggregates and rule-based events about places, not profiles of people.
5. Raw footage the project records is deleted on a fixed schedule. Retention periods are written down in advance.
6. Recording for the project happens only with consent or clear signage, and never in places where children are likely to be present.
7. Approval from the university's ethics committee is obtained before any recording of people.
8. The withdrawn datasets of Table 7.6 are never used.
9. Released models carry a model card that states intended uses and excluded uses. Identification and predictive policing are among the excluded uses.
10. Access to live feeds and logs is restricted and itself logged.

11.4 Uses worth building for

The same perception layer serves purposes that need no identification at all, and the paper should present these as its motivation.

- Search and rescue at night or at sea, where thermal imaging finds people who cannot be seen.
- Detection of small drones near airports and other sensitive sites. Drones are one of RGBT-Tiny's seven classes.
- Wildlife monitoring and the detection of poaching.
- Traffic safety in fog and darkness.
- Perimeter safety on industrial sites.
- Disaster response from drones.

11.5 Draft ethics and dual-use statement

The paper will include a statement along the following lines.

This work develops a detector for small objects in visible and thermal video. It does not perform face recognition, re-identification or any form of biometric identification, and it was not evaluated for such use. All datasets were used under their research licences. No dataset withdrawn for privacy reasons was used. Footage recorded by the authors was collected with the approval of the institutional ethics committee, with faces blurred at capture. We recognise that detection and tracking can be components of surveillance systems. We therefore release the model with a card that excludes identification and predictive policing, and we encourage deployments that comply with the EU AI Act and India's Digital Personal Data Protection Act.

Risks, Timeline and Publication Strategy

This chapter turns the plan into a schedule. It lists the risks and how each will be handled. It then sets the timeline and chooses target journals, anticipates what reviewers will ask, and outlines the manuscript.

12.1 Risk register

Table 12.1 lists the ten risks most likely to delay or weaken the work, with the response planned for each.

12.2 Timeline

The schedule assumes work begins in October 2026 and targets submission in July 2027 (Figure 12.1). Five milestones mark the exit tests of the phases.

12.3 Target journals

Table 12.2 lists suitable journals with their latest reported impact factors, which must be confirmed in Journal Citation Reports before submission. The work can support up to three papers:

TABLE 12.1. Risk register. Likelihood and impact are rated low, medium or high.

Risk	Likelihood Impact		Mitigation
RGBT-Tiny access is delayed	medium	high	request on day one; begin Stage B on VT-VOD50 and UVT-VOD2024
mamba_ssm fails on T4	medium	medium	pure-PyTorch scan; sequences here are short enough
Kaggle quota is insufficient	medium	medium	two T4s per session; university cluster or Colab as contingency; fewer single-seed ablations
H1 is refuted	medium	medium	report it; the robustness protocol and H3 still stand as contributions
a competing paper appears	medium	high	monthly prior-art check; sharpen the claim toward robustness and deployment
motion estimation fails on thermal	medium	low	learned estimator; state reset on failure
scan does not export to TensorRT	high	medium	single-step state update as plain operations; plugin as last resort
label noise in benchmarks	medium	low	audit in Notebook 03; report on corrected labels where available
overfitting to one benchmark	medium	medium	cross-dataset evaluation on four further datasets
schedule slip	high	medium	each phase is a usable deliverable; Phase 4 can be dropped



FIGURE 12.1. Timeline from October 2026 to July 2027. Grey lines group the work by phase, and red diamonds mark the five milestones. Writing begins while the live system is built, so that the method and results sections are drafted as soon as the hypotheses are answered.

- *Paper A*, the method and the robustness protocol. It is the main paper and suits IEEE TCSVT, Pattern Recognition or Information Fusion.
- *Paper B*, the deployed real-time system with edge measurements. It suits the Journal of Real-Time Image Processing (Springer) or Engineering Applications of Artificial Intelligence.
- *Paper C*, an extended version for remote sensing, if the drone results are strong. It suits IEEE TGRS or the International Journal of Remote Sensing (Taylor & Francis).

TABLE 12.2. Candidate journals. Impact factors are as reported by secondary sources for the 2025 release and are marked unverified.

Journal	Publisher	JIF [†]	Fit
Information Fusion	Elsevier	15.5	Paper A if framed around fusion robustness
IEEE Trans. Image Processing	IEEE	13.7	Paper A with strong analysis; demanding
IEEE Trans. Circuits Syst. Video Technol.	IEEE	11.1	Paper A, best fit: video and streaming
IEEE Trans. Multimedia	IEEE	9.7	Paper A; published Fusion-Mamba
IEEE Trans. Geosci. Remote Sensing	IEEE	8.6	Paper C
Eng. Appl. of Artificial Intelligence	Elsevier	8.0	Paper B
Pattern Recognition	Elsevier	7.6	Paper A
Neural Networks	Elsevier	6.3	state-space analysis
Neural Computing and Applications	Springer	4.5–6	fallback for A
Computer Vision and Image Understanding	Elsevier	3.6	published UAVDet
J. Real-Time Image Processing	Springer	3.0	Paper B, best fit
The Visual Computer	Springer	2.9	fallback
Int. J. Remote Sensing	Taylor & Francis	2.6	Paper C
Machine Vision and Applications	Springer	2.3	fallback

12.4 What reviewers will ask

Reviewers at these journals ask for the same things repeatedly. The protocol of Chapter 8 answers each in advance.

- At least three datasets, including a video dataset and a dataset with real misalignment.

- Comparisons against 2025–2026 methods, not only against YOLOv8: WaveMamba, COMO, MambaST and the alignment-free video detectors.
- Parameters, GFLOPs and latency for every method, measured on the same GPU at batch one, end to end. Jetson numbers are needed if real time is claimed.
- Mean and standard deviation over at least three seeds, with confidence intervals that respect the correlation of frames within sequences.
- An ablation for every component, failure cases and visualisations of what the state holds.
- Code, weights and a data-availability statement.

12.5 Manuscript outline

Table 12.3 gives the planned structure of Paper A. The method and experiments sections take most of the space, as reviewers expect.

TABLE 12.3. Outline of the main manuscript (Paper A), with approximate lengths for a journal of the IEEE Transactions format.

Section	Content	Pages
Introduction	problem, gap, contributions, figure of the streaming setting	1.25
Related work	YOLO and SSM detectors; RGB–T fusion; temporal and streaming detection	1.25
Method	CMFM with gated step; TSM; motion compensation; training	3.0
Robustness protocol	perturbations, severities, event metrics	0.75
Experiments	datasets, baselines, main results, ablations, robustness, efficiency	4.0
Analysis	what the state holds, failure cases, limitations	1.0
Conclusion and statements	conclusion, ethics, data and code availability	0.5

12.6 Before submission

Five checks come immediately before submission.

1. Prior art is searched again on arXiv and in the target journals across the four lanes of Figure 5.1. The claims are adjusted to whatever has appeared.
2. The code is released under the AGPL-3.0 licence that Ultralytics requires, with configurations and the run logs behind every table.

3. Weights and the robustness protocol's corruption seeds are released, so that others can evaluate on identical streams.
4. The ethics and data statements of Chapter 11 are completed.
5. A colleague who has not seen the project reproduces one main table from the released material.

Notation

The symbols below are used in Chapters 2 and 6. Superscripts v and θ always denote the visible and thermal modalities; τ always denotes time.

TABLE A.1. Notation.

Symbol	Meaning
I_t^v, I_t^θ	visible and thermal frames at time t
P_l	feature level at stride 2^l
$F_t^{v,l}, F_t^{\theta,l}$	backbone features of each modality at level l
$\tilde{F}^\theta$	thermal features resampled onto the visible grid
o	displacement field predicted for resampling
r^v, r^θ	per-location reliability of each modality, in $(0, 1)$
s_t^v, s_t^θ	sensor availability flags from the stream monitor
ρ_t	combined reliability, $\max(r^v, r^\theta)$
Z_t^l	fused features from the CMFM block
h_k, S_t	state of a selective scan; state of the Temporal State Memory
A, B, C, D	state-space parameters; A is diagonal and negative
Δ	discretisation step; small values hold the state, large values overwrite it
$\bar{A}, \bar{B}$	discretised parameters under the zero-order hold
N	state size per channel
C, C', C_s	channel counts of features, projected tokens and TSM state
τ_t, τ_0	measured frame interval and nominal training interval
$H_{t-1 \rightarrow t}$	estimated camera motion between consecutive frames

Symbol	Meaning
$\mathcal{W}, \mathcal{S}$	warp of the state by a homography; resampling by a displacement field
L	clip length in Stage B training
q_t^m	reliability target from the corruption simulator
λ_{rel}	weight of the reliability loss
$\text{AP}, \text{AP}_{50}, \text{AP}_s$	COCO average precision over IoU 0.5 to 0.95, at IoU 0.5, and for small objects
$\text{HOTA}, \text{IDF1}$	tracking metrics combining detection and association; identity F1
AUDC, R_{95}	area under the degradation curve; recovery time to 95% recall

Sources and Verification

This appendix records how the evidence in the dossier was gathered, and which figures could not be confirmed against a primary source.

B.1 How the evidence was gathered

SOURCE PAPERS. The four supplied PDFs were converted to text and read in full, including their reference lists. Their tables were lost in conversion, so every results table was read again from rendered images of the original pages. Disagreements between a paper’s text and its tables are listed in Table 3.7.

LITERATURE SURVEY. Searches covered arXiv, conference proceedings (CVPR, ICCV, ECCV, NeurIPS, ICLR, ICML, AAI, ACM MM), journal sites and official repositories up to 2 October 2026. Work from 2026 was given priority. Numbers were taken from primary pages, meaning arXiv abstracts or full text, official GitHub repositories and official documentation, wherever these could be reached.

DATASETS AND TOOLS. Dataset statistics and licences were taken from official dataset pages and repositories. Software versions were taken from the Python Package Index on the dates stated.

B.2 Figures that could not be confirmed

The following items are marked with a dagger (†) in the text. They came from secondary sources or from search snippets. Each must be checked against the primary source before it is quoted in a manuscript.

- UAVDet’s reported parameter counts of 3.74M and 3.40M. They are printed in the paper but are inconsistent with its other tables.
- CFMW’s LLVIP and SWVI mAP values, which differ between the tables and the text of the paper.
- The COCO AP of DINOv3 7B with Plain-DETR (65.6).
- EFSI-DETR’s VisDrone results (33.1 AP, 24.8 AP_S).
- The DanceTrack HOTA of MOTRv2, OC-SORT and ByteTrack, and the SportsMOT HOTA of the SAM 3 mask-propagation method.
- The SODA size thresholds.
- The licence of DroneVehicle.
- Kaggle’s weekly GPU quota.
- The postponement of the EU AI Act’s high-risk obligations to December 2027.
- The commencement dates of India’s DPDP Rules.
- The journal impact factors in Table 12.2.

B.3 Statements of absence

Several conclusions in the dossier rest on the *absence* of prior work. Examples are the absence of a causal, constant-memory state-space detector for visible–thermal streams, of a benchmark protocol for temporal sensor failure, and of a native TensorRT operator for two-dimensional selective scanning. These statements mean only that the searches did not find such work. The field publishes rapidly, and two connected literature services could not be queried because they required authorisation. The searches must therefore be repeated before submission, as Section 12.6 requires.

Bibliography

Identifiers for preprints were taken from arXiv listing and abstract pages accessed between 1 and 2 October 2026. Where an author list could not be confirmed, the work is identified by its name, venue and identifier.

- [1] A2Mamba. Vision backbone combining attention with state-space mixing. *arXiv preprint arXiv:2507.16624*, 2025.
- [2] N. Aharon, R. Orfaig, and B.-Z. Bobrovsky. BoT-SORT: Robust associations multi-pedestrian tracking. *arXiv preprint arXiv:2206.14651*, 2022.
- [3] AKCMamba-YOLO. Channel-interaction and cross-layer Mamba modules for YOLO detectors. In *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2026.
- [4] F. C. Akyon, S. O. Altinuc, and A. Temizel. Slicing aided hyper inference and fine-tuning for small object detection. In *Proc. IEEE Int. Conf. Image Processing (ICIP)*, 2022.
- [5] N. Ballas, L. Yao, C. Pal, and A. Courville. Delving deeper into convolutional networks for learning video representations. In *Proc. Int. Conf. Learning Representations (ICLR)*, 2016.
- [6] J. Cao, J. Pang, X. Weng, R. Khrodkar, and K. Kitani. Observation-centric SORT: Rethinking SORT for robust multi-object tracking. In *Proc. CVPR*, 2023.
- [7] Position-shift-robust visible–infrared detection with the CBC and SWCA modules. *arXiv preprint arXiv:2502.09311*, 2025.
- [8] CFGPNet. Visible–infrared object detection. *arXiv preprint arXiv:2608.06205*, 2026.
- [9] C-GSPN. Scaling parallel sequence models for vision. *arXiv preprint arXiv:2606.00746*, 2026.
- [10] Contextually-guided state space fusion for misaligned multispectral object detection. In *Proc. ACM Int. Conf. Multimedia (ACM MM)*, 2025. doi:10.1145/3746027.3754550.
- [11] Y.-T. Chen, J. Shi, Z. Ye, C. Mertz, D. Ramanan, and S. Kong. Multimodal object detection via probabilistic ensembling. In *Proc. European Conf. Computer Vision (ECCV)*, 2022.
- [12] J. Chen, S.-H. Kao, H. He, W. Zhuo, S. Wen, C.-H. Lee, and S.-H. G. Chan. Run, don’t walk: Chasing higher FLOPS for faster neural networks. In *Proc. CVPR*, 2023.

- [13] G. Cheng, X. Yuan, X. Yao, K. Yan, Q. Zeng, X. Xie, and J. Han. Towards large-scale small object detection: Survey and benchmarks. *IEEE Trans. Pattern Analysis and Machine Intelligence*, 2023.
- [14] COMO: Cross-Mamba interaction and offset-guided fusion for multimodal object detection. *Information Fusion*, vol. 125, 2026. *arXiv preprint* arXiv:2412.18076, 2024.
- [15] K. Corona, K. Osterdahl, R. Collins, and A. Hoogs. MEVA: A large-scale multiview, multimodal video dataset for activity detection. In *Proc. IEEE/CVF Winter Conf. Applications of Computer Vision (WACV)*, 2021.
- [16] Y. Cui, C. Zeng, X. Zhao, Y. Yang, G. Wu, and L. Wang. SportsMOT: A large multi-object tracking dataset in multiple sports scenes. In *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2023.
- [17] DAMamba: Vision state space model with dynamic adaptive scan. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2502.12627.
- [18] T. Dao. FlashAttention-2: Faster attention with better parallelism and work partitioning. *arXiv preprint* arXiv:2307.08691, 2023.
- [19] T. Dao and A. Gu. Transformers are SSMS: Generalized models and efficient algorithms through structured state space duality. In *Proc. Int. Conf. Machine Learning (ICML)*, 2024.
- [20] A. Dave, T. Khurana, P. Tokmakov, C. Schmid, and D. Ramanan. TAO: A large-scale benchmark for tracking any object. In *Proc. ECCV*, 2020.
- [21] DCHNet and the DVT-VOD1000 benchmark for unaligned visible–thermal video object detection. *arXiv preprint* arXiv:2607.08191, 2026.
- [22] NVIDIA. DeepStream SDK 9.1: Documentation and release notes, 2026. <https://docs.nvidia.com/metropolis/deepstream>.
- [23] DEIMv2: Real-time object detection meets DINOv3. *arXiv preprint* arXiv:2509.20787, 2025.
- [24] P. Dendorfer, H. Rezatofighi, A. Milan, J. Shi, D. Cremers, I. Reid, S. Roth, K. Schindler, and L. Leal-Taixé. MOT20: A benchmark for multi object tracking in crowded scenes. *arXiv preprint* arXiv:2003.09003, 2020.
- [25] DERNet. Frequency-domain small-object detection. *arXiv preprint* arXiv:2606.23825, 2026.
- [26] J. Ding, N. Xue, G.-S. Xia, et al. Object detection in aerial images: A large-scale benchmark and challenges. *IEEE Trans. Pattern Analysis and Machine Intelligence*, 2021.
- [27] DLRMamba. Low-rank selective scanning with distillation for edge multispectral detection. *arXiv preprint* arXiv:2603.06920, 2026.
- [28] Dome-DETR. Tiny object detection with density-oriented feature-query manipulation. In *Proc. ACM MM*, 2025. arXiv:2505.05741.
- [29] W. Dong et al. Fusion-Mamba for cross-modality object detection. *IEEE Trans. Multimedia*, vol. 27, 2025. arXiv:2404.09146.
- [30] Government of India. The Digital Personal Data Protection Act, 2023 (No. 22 of 2023).
- [31] Ministry of Electronics and Information Technology, Government of India. Digital Personal Data Protection Rules, 2025. Notified November 2025.

- [32] DRGBT-1K. Visible–thermal tracking benchmark with modality and platform switching. *arXiv preprint* arXiv:2607.19772, 2026.
- [33] D. Du, Y. Qi, H. Yu, Y. Yang, K. Duan, G. Li, W. Zhang, Q. Huang, and Q. Tian. The unmanned aerial vehicle benchmark: Object detection and tracking. In *Proc. ECCV*, 2018.
- [34] EFSI-DETR. Real-time small-object detection transformer for UAV imagery. *arXiv preprint* arXiv:2601.18597, 2026.
- [35] Erasure-based interaction network for RGBT video object detection and a unified benchmark (EINet, VT-VOD50). *arXiv preprint* arXiv:2308.01630, 2023.
- [36] European Parliament and Council of the European Union. Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). *Official Journal of the European Union*, 12 July 2024.
- [37] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman. The PASCAL visual object classes (VOC) challenge. *Int. J. Computer Vision*, 88(2), 2010.
- [38] H. Fan, L. Lin, F. Yang, P. Chu, G. Deng, S. Yu, H. Bai, Y. Xu, C. Liao, and H. Ling. LaSOT: A high-quality benchmark for large-scale single object tracking. In *Proc. CVPR*, 2019.
- [39] faster-coco-eval: A faster drop-in replacement for the COCO evaluation tools. https://github.com/MiXaiLL76/faster_coco_eval.
- [40] Gradio team. FastRTC: Real-time communication library for Python. <https://github.com/gradio-app/fastrtc>.
- [41] R. Gao, J. Qi, and L. Wang. Multiple object tracking as ID prediction. In *Proc. CVPR*, 2025. arXiv:2403.16848.
- [42] European Parliament and Council of the European Union. Regulation (EU) 2016/679 (General Data Protection Regulation). *Official Journal of the European Union*, L 119, 2016.
- [43] M. Gehrig and D. Scaramuzza. Recurrent vision transformers for object detection with event cameras. In *Proc. CVPR*, 2023.
- [44] A. González, Z. Fang, Y. Socarras, J. Serrat, D. Vázquez, J. Xu, and A. M. López. Pedestrian detection at day/night time with visible and FIR cameras: A comparison. *Sensors*, 16(6), 2016.
- [45] Gradio: Build machine learning web applications in Python. Version 6.29, 2026. <https://github.com/gradio-app/gradio>.
- [46] GraphScan. Graph-based scanning for vision state space models. *arXiv preprint* arXiv:2605.11300, 2026.
- [47] A. Gu, K. Goel, and C. Ré. Efficiently modeling long sequences with structured state spaces. In *Proc. ICLR*, 2022.
- [48] A. Gu and T. Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint* arXiv:2312.00752, 2023.
- [49] A. Gupta, P. Dollár, and R. Girshick. LVIS: A dataset for large vocabulary instance segmentation. In *Proc. CVPR*, 2019.

- [50] D. Han et al. Demystify Mamba in vision: A linear attention perspective. In *Advances in NeurIPS*, 2024. arXiv:2405.16605.
- [51] A. Hatamizadeh and J. Kautz. MambaVision: A hybrid Mamba–Transformer vision backbone. In *Proc. CVPR*, 2025. arXiv:2407.08083.
- [52] HDMamba-YOLO. Stage-wise mixed SSM–CNN detector for small objects. *arXiv preprint* arXiv:2609.23061, 2026.
- [53] J.-Y. He et al. DAMO-StreamNet: Optimizing streaming perception in autonomous driving. In *Proc. Int. Joint Conf. Artificial Intelligence (IJCAI)*, 2023.
- [54] HGT-Track and the VT-Tiny-MOT benchmark for visible–thermal tiny multi-object tracking. *arXiv preprint* arXiv:2412.10861, 2024.
- [55] J. Ho, A. Jain, and P. Abbeel. Denoising diffusion probabilistic models. In *Advances in NeurIPS*, 2020.
- [56] Y.-X. Huang et al. DQ-DETR: DETR with dynamic query for tiny object detection. In *Proc. ECCV*, 2024.
- [57] T. Huang et al. LocalMamba: Visual state space model with windowed selective scan. *arXiv preprint* arXiv:2403.09338, 2024.
- [58] S. Huang et al. DEIM: DETR with improved matching for fast convergence. In *Proc. CVPR*, 2025. arXiv:2412.04234.
- [59] S. Hwang, J. Park, N. Kim, Y. Choi, and I. S. Kweon. Multispectral pedestrian detection: Benchmark dataset and baseline. In *Proc. CVPR*, 2015.
- [60] X. Jia, C. Zhu, M. Li, W. Tang, and W. Zhou. LLVIP: A visible-infrared paired dataset for low-light vision. In *Proc. ICCV Workshops*, 2021.
- [61] M. Kang, C.-M. Ting, F. F. Ting, and R. C.-W. Phan. ASF-YOLO: A novel YOLO model with attentional scale sequence fusion for cell instance segmentation. *Image and Vision Computing*, 2024.
- [62] A. Kuznetsova et al. The Open Images dataset V4. *Int. J. Computer Vision*, 128, 2020.
- [63] D. Lam et al. xView: Objects in context in overhead imagery. *arXiv preprint* arXiv:1802.07856, 2018.
- [64] LCMamNet. Lightweight Mamba network for infrared small targets, deployed on Jetson Orin NX. *arXiv preprint* arXiv:2607.24184, 2026.
- [65] M. Lei et al. YOLOv13: Real-time object detection with hypergraph-enhanced adaptive visual perception. *arXiv preprint* arXiv:2506.17733, 2025.
- [66] C. Li, D. Song, R. Tong, and M. Tang. Multispectral pedestrian detection via simultaneous detection and segmentation. In *Proc. British Machine Vision Conf. (BMVC)*, 2018. arXiv:1808.04818.
- [67] M. Li, Y.-X. Wang, and D. Ramanan. Towards streaming perception. In *Proc. ECCV*, 2020.
- [68] C. Li et al. LongShortNet: Exploring temporal and semantic features fusion in streaming perception. In *Proc. IEEE ICASSP*, 2023.

- [69] H. Li, Q. Hu, B. Zhou, Y. Yao, J. Lin, K. Yang, and P. Chen. CFMW: Cross-modality fusion Mamba for robust object detection under adverse weather. *IEEE Trans. Circuits and Systems for Video Technology*, 35(12):12066–12081, 2025. doi:10.1109/TCSVT.2025.3587918.
- [70] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick. Microsoft COCO: Common objects in context. In *Proc. ECCV*, 2014.
- [71] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie. Feature pyramid networks for object detection. In *Proc. CVPR*, 2017.
- [72] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia. Path aggregation network for instance segmentation. In *Proc. CVPR*, 2018.
- [73] J. Liu, X. Fan, Z. Huang, G. Wu, R. Liu, W. Zhong, and Z. Luo. Target-aware dual adversarial learning and a multi-scenario multi-modality benchmark to fuse infrared and visible for object detection. In *Proc. CVPR*, 2022.
- [74] W. Liu, H. Lu, H. Fu, and Z. Cao. Learning to upsample by learning to sample. In *Proc. ICCV*, 2023.
- [75] X. Liu, H. Peng, N. Zheng, Y. Yang, H. Hu, and Y. Yuan. EfficientViT: Memory efficient vision transformer with cascaded group attention. In *Proc. CVPR*, 2023.
- [76] Y. Liu, Y. Tian, Y. Zhao, H. Yu, L. Xie, Y. Wang, Q. Ye, and Y. Liu. VMamba: Visual state space model. In *Advances in NeurIPS*, 2024. arXiv:2401.10166.
- [77] J. Luiten, A. Ošep, P. Dendorfer, P. Torr, A. Geiger, L. Leal-Taixé, and B. Leibe. HOTA: A higher order metric for evaluating multi-object tracking. *Int. J. Computer Vision*, 129, 2021.
- [78] C. Lyu, W. Zhang, H. Huang, Y. Zhou, Y. Wang, Y. Liu, S. Zhang, and K. Chen. RTMDet: An empirical study of designing real-time object detectors. *arXiv preprint arXiv:2212.07784*, 2022.
- [79] M3OT: A multi-drone visible–infrared multi-object tracking dataset. *Scientific Data*, 2025. doi:10.1038/s41597-025-06204-0.
- [80] S. Ma and Y. Xu. MPDIoU: A loss for efficient and accurate bounding box regression. *arXiv preprint arXiv:2307.07662*, 2023.
- [81] MambaNeXt-YOLO. Hybrid CNN–Mamba detector for embedded devices. *arXiv preprint arXiv:2506.03654*, 2025.
- [82] MambaPSA: A Mamba replacement for C2PSA in YOLO26. *arXiv preprint arXiv:2607.12681*, 2026.
- [83] A. Torres-Legueta. mamba.py: A simple and efficient Mamba implementation in pure PyTorch. <https://github.com/alxndrTL/mamba.py>.
- [84] MambaRefine-YOLO. Dual-gated Mamba fusion for visible–infrared small-object detection. *arXiv preprint arXiv:2511.19134*, 2025.
- [85] State Spaces. mamba: Official implementation of Mamba and Mamba-2. <https://github.com/state-spaces/mamba>.
- [86] MambaST: A plug-and-play cross-spectral spatial-temporal fuser for efficient pedestrian detection. In *Proc. IEEE Int. Conf. Intelligent Transportation Systems (ITSC)*, 2024. arXiv:2408.01037.

- [87] MambaTrack. Mamba-based motion prediction for multi-object tracking. In *Proc. ACM MM*, 2024. arXiv:2408.09178.
- [88] MATR. End-to-end multi-object tracking. *arXiv preprint* arXiv:2509.21715, 2025.
- [89] A. Milan, L. Leal-Taixé, I. Reid, S. Roth, and K. Schindler. MOT16: A benchmark for multi-object tracking. *arXiv preprint* arXiv:1603.00831, 2016.
- [90] MMOT: A multispectral multi-object tracking benchmark. In *NeurIPS Datasets and Benchmarks Track*, 2025. arXiv:2510.12565.
- [91] MS2Fusion. Shared- and cross-parameter state-space fusion for multispectral object detection. *Information Fusion*, 2025. arXiv:2507.14643.
- [92] MSDf-Mamba. Deformable Mamba fusion correcting visible–infrared misalignment. *Remote Sensing*, 17(24), 2025.
- [93] MSeNet and the UVT-VOD2024 benchmark for alignment-free visible–thermal video object detection. *arXiv preprint* arXiv:2410.12143, 2024.
- [94] MSGNet. Spatio-temporal graph network for alignment-free visible–thermal video object detection. *arXiv preprint* arXiv:2504.11779, 2025.
- [95] O2-DEIM and O2-DFINE. Real-time oriented detection transformers. *IEEE Trans. Geoscience and Remote Sensing*, 2026. arXiv:2603.15497.
- [96] ObjM: Objectness-guided scanning for efficient vision Mamba. *Knowledge-Based Systems*, 2025.
- [97] M. Oquab et al. DINOv2: Learning robust visual features without supervision. *Trans. Machine Learning Research*, 2024.
- [98] O. Özdenizci and R. Legenstein. Restoring vision in adverse weather conditions with patch-based denoising diffusion models. *IEEE Trans. Pattern Analysis and Machine Intelligence*, 45(8), 2023.
- [99] K. Peng, A. Mathur, and A. Narayanan. Mitigating dataset harms requires stewardship: Lessons from 1000 papers. In *NeurIPS Datasets and Benchmarks Track*, 2021. arXiv:2108.02922.
- [100] Y. Peng et al. D-FINE: Redefine regression task in DETRs as fine-grained distribution refinement. In *Proc. ICLR*, 2025. arXiv:2410.13842.
- [101] J. Nolan, creator. *Person of Interest*. Television series, CBS, 2011–2016.
- [102] PRISMamba. Partial ring scan for vision state space models. In *Proc. ICML*, 2026. arXiv:2602.04170.
- [103] Supreme Court of India. Justice K. S. Puttaswamy (Retd.) and Anr. v. Union of India and Ors., (2017) 10 SCC 1.
- [104] PyAV: Pythonic bindings for FFmpeg. Version 19.0, 2026. <https://github.com/PyAV-Org/PyAV>.
- [105] F. Qingyun, H. Dapeng, and W. Zhaokui. Cross-modality fusion transformer for multispectral object detection. *arXiv preprint* arXiv:2111.00273, 2021.

- [106] S. Razakarivony and F. Jurie. Vehicle detection in aerial imagery: A small target detection benchmark. *J. Visual Communication and Image Representation*, 34, 2016.
- [107] RegisterBridgeMM. Visible–infrared detection with a frozen DINOv3 encoder. *arXiv preprint arXiv:2608.04833*, 2026.
- [108] RemoteDet-Mamba. Siamese CNN with quad-directional cross-modal Mamba for remote sensing detection. In *Proc. IEEE ICASSP*, 2026. arXiv:2410.13532.
- [109] E. Ristani, F. Solera, R. Zou, R. Cucchiara, and C. Tomasi. Performance measures and a data set for multi-target, multi-camera tracking. In *Proc. ECCV Workshops*, 2016.
- [110] P. Robicheaux et al. Roboflow100-VL: A multi-domain object detection benchmark for vision-language models. *arXiv preprint arXiv:2505.20612*, 2025.
- [111] I. Robinson et al. RF-DETR: Neural architecture search for real-time detection transformers. In *Proc. ICLR*, 2026. arXiv:2511.09554.
- [112] Roboflow. trackers: Multi-object tracking algorithms under the Apache licence. <https://github.com/roboflow/trackers>.
- [113] RT-DETRv4. Real-time detection transformer distilled from DINOv3. *arXiv preprint arXiv:2510.25257*, 2025.
- [114] O. Russakovsky et al. ImageNet large scale visual recognition challenge. *Int. J. Computer Vision*, 115, 2015.
- [115] Meta AI. SAM 3: Segment anything with concepts. *arXiv preprint arXiv:2511.16719*, 2025.
- [116] SCDT. Visible–thermal tracking under missing modalities. In *Proc. CVPR*, 2026. arXiv:2607.24701.
- [117] ScopeMamba-YOLO. Off-path selective scanning for small-object detection. *arXiv preprint arXiv:2609.10156*, 2026.
- [118] M. Segu, L. Piccinelli, S. Li, Y.-H. Yang, B. Schiele, and L. Van Gool. Samba: Synchronized set-of-sequences modeling for multiple object tracking. In *Proc. ICLR*, 2025. arXiv:2410.01806.
- [119] SFDNet. Spectrum-aware tiny-object detection. In *Proc. ECCV*, 2026. arXiv:2606.29029.
- [120] SF-Mamba. Efficient vision Mamba at short token lengths. *arXiv preprint arXiv:2603.16423*, 2026.
- [121] S. Shao, Z. Li, T. Zhang, C. Peng, G. Yu, X. Zhang, J. Li, and J. Sun. Objects365: A large-scale, high-quality dataset for object detection. In *Proc. ICCV*, 2019.
- [122] J. Shen et al. ICAFusion: Iterative cross-attention guided feature fusion for multispectral object detection. *Pattern Recognition*, 2024. arXiv:2308.07504.
- [123] K. Shim et al. Focusing on tracks for online multi-object tracking. In *Proc. CVPR*, 2025.
- [124] O. Siméoni et al. DINOv3. *arXiv preprint arXiv:2508.10104*, 2025.
- [125] S. Singh et al. Benchmarking object detectors with COCO: A new path forward. In *Proc. ECCV*, 2024.

- [126] J. T. H. Smith, A. Warrington, and S. W. Linderman. Simplified state space layers for sequence modeling. In *Proc. ICLR*, 2023.
- [127] Selective mask propagation with SAM 3 for multi-object tracking. *arXiv preprint arXiv:2606.13033*, 2026.
- [128] J. Song, C. Meng, and S. Ermon. Denoising diffusion implicit models. In *Proc. ICLR*, 2021.
- [129] SpikingMOT. Spiking neural networks for multi-object tracking. *arXiv preprint arXiv:2607.19875*, 2026.
- [130] P. Sun, J. Cao, Y. Jiang, Z. Yuan, S. Bai, K. Kitani, and P. Luo. DanceTrack: Multi-object tracking in uniform appearance and diverse motion. In *Proc. CVPR*, 2022.
- [131] Y. Sun, B. Cao, P. Zhu, and Q. Hu. Drone-based RGB-infrared cross-modality vehicle detection via uncertainty-aware learning. *IEEE Trans. Circuits and Systems for Video Technology*, 32(10), 2022.
- [132] Y. Tian, Q. Ye, and D. Doermann. YOLOv12: Attention-centric real-time object detectors. In *Advances in NeurIPS*, 2025. *arXiv:2502.12524*.
- [133] TinySet-9M. Multi-domain small-object detection dataset. *arXiv preprint arXiv:2604.02773*, 2026.
- [134] TMambaDet. Temporal Mamba for video object detection. In *Proc. CVPR*, 2026.
- [135] TOLF. Tiny-object localisation with normalising flows. *arXiv preprint arXiv:2601.00617*, 2026.
- [136] J. Luiten and A. Hoffhues. TrackEval: Evaluation code for multi-object tracking. <https://github.com/JonathonLuiten/TrackEval>.
- [137] TrackSSM. State-space motion model for multi-object tracking. *arXiv preprint arXiv:2409.00487*, 2024.
- [138] UAV-DETR. Detection transformer for UAV imagery. *arXiv preprint arXiv:2501.01855*, 2025.
- [139] L. A. Varga, B. Kiefer, M. Messmer, and A. Zell. SeaDronesSee: A maritime benchmark for detecting humans in open water. In *Proc. WACV*, 2022.
- [140] VNCT: Vision non-causal trapezoidal Mamba. *arXiv preprint arXiv:2607.03589*, 2026.
- [141] Visible-thermal multiple object tracking: Large-scale video dataset and progressive fusion approach (VT-MOT). *Pattern Recognition; arXiv preprint arXiv:2408.00969*, 2024.
- [142] J. Wang, W. Yang, H. Guo, R. Zhang, and G.-S. Xia. Tiny object detection in aerial images. In *Proc. Int. Conf. Pattern Recognition (ICPR)*, 2020.
- [143] J. Wang, C. Xu, W. Yang, and L. Yu. A normalized Gaussian Wasserstein distance for tiny object detection. *arXiv preprint arXiv:2110.13389*, 2021.
- [144] A. Wang, H. Chen, L. Liu, K. Chen, Z. Lin, J. Han, and G. Ding. YOLOv10: Real-time end-to-end object detection. In *Advances in NeurIPS*, 2024. *arXiv:2405.14458*.
- [145] X. Wang and H. Chen. HPS-DETR: Enhancing small object detection with lightweight feature extraction and transformer integration. *IEEE Trans. Geoscience and Remote Sensing*, 63:5937420, 2025. doi:10.1109/TGRS.2025.3583418.

- [146] Z. Wang et al. Mamba YOLO: A simple baseline for object detection with state space model. In *Proc. AAAI Conf. Artificial Intelligence*, 2025. arXiv:2406.05835.
- [147] A. Wang et al. YOLOE: Real-time seeing anything. In *Proc. ICCV*, 2025. arXiv:2503.07465.
- [148] WaveMamba: Wavelet-driven Mamba fusion for RGB-infrared object detection. In *Proc. ICCV*, 2025. arXiv:2507.18173.
- [149] R.J. Williams and J. Peng. An efficient gradient-based algorithm for on-line training of recurrent network trajectories. *Neural Computation*, 2(4), 1990.
- [150] G.-S. Xia et al. DOTA: A large-scale dataset for object detection in aerial images. In *Proc. CVPR*, 2018.
- [151] C. Xiao et al. Spatial-Mamba: Effective visual state space models via structure-aware state fusion. In *Proc. ICLR*, 2025. arXiv:2410.15091.
- [152] F. Xie et al. QuadMamba: Learning quadtree-based selective scan for visual state space model. In *Advances in NeurIPS*, 2024.
- [153] XS-VID: An extremely small video object detection dataset. *arXiv preprint* arXiv:2407.18137, 2024.
- [154] C. Xu, J. Wang, W. Yang, H. Yu, L. Yu, and G.-S. Xia. Detecting tiny objects in aerial images: A normalized Wasserstein distance and a new benchmark. *ISPRS J. Photogrammetry and Remote Sensing*, 2022.
- [155] J. Yang, S. Liu, Z. Li, X. Li, and J. Sun. Real-time object detection for streaming perception. In *Proc. CVPR*, 2022.
- [156] Y. Yang, F. Guo, and P. Niu. UAVDet: A CNN–Mamba hybrid network for efficient small object detection in UAV imagery. *Computer Vision and Image Understanding*, 264:104637, 2026. doi:10.1016/j.cviu.2026.104637.
- [157] X. Ying et al. Visible-thermal tiny object detection: A benchmark dataset and baselines. *IEEE Trans. Pattern Analysis and Machine Intelligence*, 2025. arXiv:2406.14482.
- [158] G. Jocher and J. Qiu. Ultralytics YOLO11. Software, 2024. <https://github.com/ultralytics/ultralytics>.
- [159] YOLO12-MambaScan. YOLO12 with high-frequency convolutions and Mamba context. *arXiv preprint* arXiv:2609.13647, 2026.
- [160] Ultralytics. YOLO26. Released January 2026; technical report *arXiv preprint* arXiv:2606.03748, 2026. <https://docs.ultralytics.com/models/yolo26>.
- [161] F. Yu, H. Chen, X. Wang, W. Xian, Y. Chen, F. Liu, V. Madhavan, and T. Darrell. BDD100K: A diverse driving dataset for heterogeneous multitask learning. In *Proc. CVPR*, 2020.
- [162] X. Yu, Y. Gong, N. Jiang, Q. Ye, and Z. Han. Scale match for tiny person detection. In *Proc. WACV*, 2020.
- [163] W. Yu and X. Wang. MambaOut: Do we really need Mamba for vision? In *Proc. CVPR*, 2025. arXiv:2405.07992.

- [164] X. Yuan et al. Strip R-CNN: Large strip convolution for remote sensing object detection. In *Proc. AAAI*, 2026. arXiv:2501.03775.
- [165] H. Zhang, E. Fromont, S. Lefèvre, and B. Avignon. Multispectral fusion for object detection with cyclic fuse-and-refine blocks. In *Proc. ICIP*, 2020.
- [166] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, and X. Wang. ByteTrack: Multi-object tracking by associating every detection box. In *Proc. ECCV*, 2022.
- [167] H. Zhang, C. Xu, and S. Zhang. Inner-IoU: More effective intersection over union loss with auxiliary bounding box. *arXiv preprint arXiv:2311.02877*, 2023.
- [168] Y. Zhang, T. Wang, and X. Zhang. MOTRv2: Bootstrapping end-to-end multi-object tracking by pretrained object detectors. In *Proc. CVPR*, 2023.
- [169] Y. Zhao, W. Lv, S. Xu, J. Wei, G. Wang, Q. Dang, Y. Liu, and J. Chen. DETRs beat YOLOs on real-time object detection. In *Proc. CVPR*, 2024.
- [170] X. Zhu, S. Lyu, X. Wang, and Q. Zhao. TPH-YOLOv5: Improved YOLOv5 based on transformer prediction head for object detection on drone-captured scenarios. In *Proc. ICCV Workshops*, 2021.
- [171] P. Zhu, L. Wen, D. Du, X. Bian, H. Fan, Q. Hu, and H. Ling. Detection and tracking meet drones challenge. *IEEE Trans. Pattern Analysis and Machine Intelligence*, 44(11), 2022.
- [172] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang. Vision Mamba: Efficient visual representation learning with bidirectional state space model. In *Proc. ICML*, 2024. arXiv:2401.09417.
- [173] Z. Zong, G. Song, and Y. Liu. DETRs with collaborative hybrid assignments training. In *Proc. ICCV*, 2023.
- [174] N. Zubić, M. Gehrig, and D. Scaramuzza. State space models for event cameras. In *Proc. CVPR*, 2024. arXiv:2402.15584.